

EXTENDED SUPPLEMENT TO ESTIMATING RUN-TO-RUN COVARIANCE ACROSS LANGUAGE MODEL BENCHMARKS

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Paper under double-blind review

This document holds the full-length versions of appendix sections that the submitted paper compresses. Each section below appears in the paper under the same heading, and the paper’s shorter text carries every number the main text cites. Nothing here changes a conclusion of the paper, and the numbers are the ones the paper reports. Cross-references to sections or appendices of the paper resolve within the paper, not within this document.

.1 COMMON STEP RULE

For each of the 125 configurations, we selected the largest released checkpoint step available for all three runs. That step is recorded in each reduced filename, although the supplement described in the analysis record omits the fetch stage’s complete configuration table. In 92 configurations, the three runs have different maximum steps. DataDecide stops the two auxiliary runs below 1B at 25 percent of the 1B compute budget and trains three complete runs only at 1B (Magnusson et al., 2025). The per-instance release contains auxiliary runs from 150M upward. The five replicated sizes used here begin at 150M rather than 4M. This availability rule explains the difference from DataDecide’s fourteen-size ladder.

In 26 configurations, one at 530M and 25 at 750M, the largest maximum step exceeds the smallest by more than a factor of 1.5. We indexed the member names of the 25 released recipe archives to recover every run’s maximum step. The index reproduces the selected step recorded for all 375 runs. At 530M the selected step averages 86.7 percent of the default run’s final step, with configuration values from 38.9 to 88.7 percent. At 750M it averages 41.5 percent, with values from 17.7 to 44.0 percent. Only 33 configurations have three runs that share a final step, and 24 of those sit at 1B.

Table 1 recomputes both estimates on subsets defined by the index. Excluding the 26 severely truncated configurations raises margin inflation to 1.276 and accuracy inflation to 1.120, and neither wild interval includes one. The 33 configurations with a shared final step give intervals too wide to separate inflation from one, and 18 of their 24 recipes contribute only a 1B configuration. We also define a configuration’s influence as $(T_c - \hat{\Lambda}^2 U_c) / \sum_{c'} U_{c'}$, which sums to zero and measures how far the configuration pulls the squared ratio. Its Spearman correlation with the selected step’s share of the default final step is -0.036 for margins and 0.001 for accuracy. Permuting that share within each size over 4,999 draws gives $p = 0.314$ and $p = 0.990$. Within the two truncated bands, which hold 25 configurations each, the correlations are -0.011 and 0.155 for margins and 0.275 and 0.232 for accuracy, and we didn’t test those band values separately. We chose these subsets after seeing the full-sample estimates, so they describe sensitivity and don’t test a stated hypothesis. The auxiliary-run contrast and the estimates by size in Table 6 give two further checks.

Table 1. Inflation estimates on subsets defined by each configuration’s checkpoint schedule, with wild cluster bootstrap-t intervals over 4,999 draws. Severe truncation means that the largest final step among the three runs exceeds the smallest by more than a factor of 1.5.

Subset	Configurations	Recipes	Margin	Accuracy
All	125	25	1.244 [1.143, 1.338]	1.078 [0.993, 1.157]
Shared final step	33	24	1.082 [0.121, 1.530]	1.129 [0.769, 1.397]
Without severe truncation	99	25	1.276 [1.114, 1.420]	1.120 [1.020, 1.208]
Severe truncation only	26	25	1.107 [0.692, 1.397]	0.977 [0.357, 1.306]
Different final steps	92	25	1.330 [1.024, 1.583]	1.070 [0.956, 1.171]

.2 RELIABILITY BY TRAIT

The item halves contain 18,840 and 18,842 items across the battery. The fetch stage creates a balanced split within each trait and records the split for reuse across runs. The planning document proposed estimating reliability on eight screening recipes before inspecting seventeen estimation recipes. We didn’t implement that separation, and Table 2 instead uses all 25 recipes. The reliability checks and the decisions based on them therefore remain exploratory.

Table 2. Cross-half reliability of run deviations across all 25 recipes. Reliability equals $\hat{\sigma}_E^2/(\hat{\sigma}_E^2 + \hat{\sigma}_{\text{noise}}^2)$ and can be negative when the estimated run component is negative. MC denotes a multiple-choice scoring format in this table.

Trait	Format	Accuracy	Margin
ARC-Easy	MC	0.602	0.927
ARC-Challenge	MC	0.203	0.584
BoolQ	yes/no	0.970	0.993
CommonsenseQA	MC	0.615	0.930
HellaSwag	cloze	0.450	0.757
MMLU	MC	0.022	0.860
OpenBookQA	MC	0.091	0.392
PIQA	cloze	−0.028	0.273
Social IQa	MC	0.416	0.888
WinoGrande	cloze	−0.102	−0.086
Battery mean		0.324	0.652

Two accuracy traits have negative reliability estimates, and two others fall below 0.10. Recomputing the per-trait information ratios gives a median of 1.608, a minimum of 1.571, and a maximum of 1.732, which reproduces the three figures the pipeline summary reported without releasing the ratios themselves. The battery-level ρ_g recomputes to 0.961 on margins and 0.734 on accuracy, against planning values of 0.61 for margins and 0.49 for accuracy at $\Lambda^2 = 1.819$. The reported matrices contain the mean variance used to reconcile those quantities.

.3 POWER CALCULATIONS

Table 3 reproduces the planning arithmetic for 85 configurations and seventeen recipe clusters. The completed analysis instead uses 125 configurations and 25 clusters. On the Λ scale, the reported linearised standard errors are 0.045 for margins and 0.038 for accuracy, compared with planning values of 0.0935 and 0.1232. The corresponding variance-ratio standard errors, 0.112 and 0.081, belong to the Λ^2 scale and aren’t comparable with the planning column.

The realised standard errors are smaller by factors of 2.1 and 3.3. Increasing the configuration count alone would predict a factor of approximately 1.21 under otherwise fixed assumptions. Interpolation of the N1 simulation grid gives dispersion at the planned $\bar{r}_E = 0.0911$ that is 1.04 times the dispersion at the observed margin value of 0.061, and 1.13 times the dispersion at the accuracy value of 0.0181. Differences in the design effect and reliability also affect this comparison, but we haven’t separated their contributions. We retain the original planning calculations rather than calculating power from the estimated effect after observing the results.

.4 DECISION CHECKS AND DEVIATIONS

The planning document lists nine checks with proposed responses. Its claimed date is September 5, 2026, but we haven’t established that date through an independent timestamp. We consequently describe the document as a reported plan and treat its checks as exploratory assessments.

A batch offset can contribute to the same cross-trait products as run-level variation. The auxiliary-run contrast removes the shared default-versus-auxiliary contrast from that estimate, while the estimate using all three runs remains exposed. Check G8 passes on point estimates, although the intervals cross the decision boundaries. We haven’t corrected the full-set estimate for this exposure.

Table 3. Planning calculations at $\Lambda^2 = 1.819$ and $N = 85$ configurations. The calculations use $t(16)$, a design effect of $1 + 4\rho_{\text{ICC}} = 1.60$ at $\rho_{\text{ICC}} = 0.15$, and $\rho_g = 0.61$ for margins or 0.49 for accuracy. Power entries are reported from the planning document, which doesn’t record the alternative hypothesis they were computed against.

Score scale	Aggregation	Planned SE(Λ)	MDE(Λ)	Power
Margin	Cross-half	0.0935	1.300	0.89
Margin	Battery mean	0.0845	n/a	0.94
Accuracy	Cross-half	0.1232	1.412	0.68
Accuracy	Battery mean	0.1076	n/a	0.79
Accuracy	Cross-half, $\bar{r}_E = 0.115$	0.1232	n/a	0.82

Under the default gain simulator, $\sigma_g = 0.05$ produces approximately 61 percent of the observed margin excess. An empirically matched simulator produces approximately eight percent at the same gain scale and six percent at the estimated scale of 0.042. These results describe sensitivity to simulator assumptions and aren’t causal bounds on the contribution of gain. The accuracy result offers no independent protection against this mechanism because multiplying a margin by a positive factor leaves its sign unchanged. Partialling the gain covariate increases Λ_M , and adding gain to the rank-one predictor worsens held-out point error relative to the scalar predictor alone, but these observations don’t identify a unique causal explanation. Table 4 records every planning check beside the outcome the source analysis recorded for it.

Table 4. Planning checks, proposed responses, and observed outcomes in the source analysis. The original assessment labels four checks as failures, while G2 uses a lower threshold that can’t discriminate under its model.

Check	Condition and timing	Proposed response	Reported outcome
G0	Outputs parse at the common step, day 1	Drop unavailable cells, report reduced N if fewer than 70 estimation cells remain	Pass, all 125 cells
G1	Accuracy power at $\bar{r}_E = 0.115$ at least 0.80, day 2	Make accuracy secondary and report margin in the abstract	Pass, power 1.00 against an inflation of 1.427 at $\rho_g = 0.73$
G2	Median information ratio at least 1.4, day 2	Remove comparative information claim	Ratio 1.61, threshold below the ratio’s floor
G3	Symmetric cross-half covariance and positive diagonal, day 3	Split by source document, then report leakage if asymmetry persists	Fail, one negative margin diagonal and two accuracy diagonals
G4	$ \hat{\Lambda} - 1 \leq 0.02$ under N1 and N2, day 4	Subtract calibrated offset and report both estimates	Pass, offset at most 0.002
G5	N5 accounts for less than 50 percent of excess, day 5	Lead with sharpness and report accuracy alone	Fail, default simulator share 0.61
G6	PolyPythias transport error within 20 percent, day 6	Report transport failure and its ratio	Fail upward under the excess-ratio reading
G7	Held-out loss available in metadata, day 1	Use leave-one-trait-out competence covariate	Fail, no required field
G8	Auxiliary-run $\Lambda \geq 1 + 0.5(\Lambda_{\text{full}} - 1)$, day 5	Lead with auxiliary-run estimate if condition fails	Pass on both point estimates, margin 1.256 versus 1.122, intervals cross their boundaries

The analysis didn’t execute the proposed responses for G3 and G5. G3 combines asymmetry with negative variance estimates, although a negative moment estimate doesn’t establish leakage. The available records also lack a usable source-document identifier for the proposed alternative split. We retain the margin analysis after the G5 failure with the simulator sensitivity disclosed, which departs from the proposed response. The source gate table contains seven rows, with transport and the auxiliary-run contrast supplying G6 and G8 separately.

162 .5 REPORTED PLANNING PREDICTIONS

163
164 The planning document specifies eleven predictions and their rejection conditions. Table 5 records
165 the outcomes without treating the document as verified preregistration.

- 166 1. Both $\Lambda_M > 1.349$ and $\Lambda_A > 1.40$, with lower confidence limits above one, constitute the
167 first planned prediction.
- 168 2. The accuracy estimate retains at least 80 percent of the margin estimate, expressed as $\Lambda_A \geq$
169 $0.8\Lambda_M$.
- 170 3. Accuracy retains more than 70 percent of its excess after gain adjustment, and N5 generates
171 less than 30 percent of the margin excess.
- 172 4. At measured token-budget ratios, N6 generates less than 50 percent of the full-set excess.
- 173 5. The independence calculation underpredicts aggregate seed standard deviation by Λ in at
174 least four of five size bands.
- 175 6. DataDecide’s R_E predicts PolyPythias aggregate seed standard deviation within 20 percent,
176 while independence misses by more than 20 percent.
- 177 7. At least one seed mechanism in the external panel gives $\Lambda > 1$ and a bootstrap lower limit
178 above one.
- 179 8. At least eight traits have an information ratio between 1.4 and 3.0 in the planned assess-
180 ment.
- 181 9. The cross-format diagnostic exceeds 1.2, with its lower confidence limit also exceeding
182 one.
- 183 10. The Frobenius distance $\|R_E - R_P\|_F$ exceeds 0.5 in the comparison between the two
184 correlation matrices.
- 185 11. More than ten percent of same-size configuration pairs have $P_{\text{flip}} > 0.05$ at the measured
186 aggregate noise scale.

191 .6 RELATION TO PRIOR MEASUREMENT WORK

192
193 Reimers and Gurevych (2017) compare score distributions across seeds within datasets, while
194 Madaan et al. (2024) report benchmark-level seed variance. Hagmann et al. (2023) model non-
195 determinism with scalar variance components and their interactions with data properties. Our target
196 adds covariance between benchmark-specific run deviations to that measurement problem.

197 Jordan (2024) attributes much test-set variance to per-example variation across seeds, and its Ap-
198 pendix C compares run accuracy across five ImageNet evaluation sets with R^2 at most 0.14. Those
199 comparisons concern shifted versions of one benchmark. Bui et al. (2025) vary fine-tuning seeds,
200 and Hochlehnert et al. (2025) vary decoding seeds, which address different variance components
201 from the pretraining-run variation examined here. Dodge et al. (2020, Section 4.3) ask whether
202 a seed performs well across datasets and compare four GLUE task rankings without estimating a
203 covariance matrix.

204 Zhou et al. (2020, Figure 3) report a Spearman correlation matrix across datasets. Their standard
205 block has correlations at least 0.95, while standard-to-analysis entries include 0.97 for STR-L and
206 0.30 for STR-O. These correlations compare checkpoints within runs and average across seeds,
207 so they combine changes in checkpoint performance rather than isolating within-configuration seed
208 deviations. Dufour et al. (2026) separate training and sampling variation in FID, while Miller (2024)
209 estimates item-level uncertainty for one model. Zhao et al. (2026) report the three-seed standard
210 deviation of a twelve-task average and combine twelve per-task Welch tests in their Appendix E.4
211 without estimating cross-task covariance.

212 Haupt et al. (2026) use the PolyPythias 410M panel to estimate per-item variation and prioritise
213 items by welfare weight and improvability. Their per-item variance estimates and approximate
214 item-independence assumption address a different quantity from the off-diagonal covariance of
215 benchmark run effects. Their item-level estimates and our benchmark covariance therefore answer
different questions using some of the same models.

Cheverud (2001), Nyholt (2004), and Li and Ji (2005) estimate effective numbers of independent tests from observed correlation matrices. Cheverud uses eigenvalue variance, Nyholt uses spectral decomposition of linkage disequilibrium, and Li and Ji revise the effective-count approximation. Their application adjusts significance thresholds, whereas ours converts an estimated noise covariance into an aggregate standard deviation and an effective benchmark count. Searle et al. (1992) explain negative variance-component estimates from moment methods when the true component lies near zero.

Schaeffer et al. (2023) show how discontinuous metrics can change the appearance of performance trends. Accuracy thresholds our margin at zero, so we report both scales while retaining their separate noise properties. Low estimated seed signal relative to item noise provides a measurement limitation for accuracy in this study. Dehghani et al. (2021) show that benchmark selection can reorder methods, which concerns the composition sensitivity examined here.

Demšar (2006) and Dror et al. (2017) develop inference across datasets, while Card et al. (2020) examine statistical power and Gorman and Bedrick (2019) examine dependence on a single split. Burnell et al. (2023) advocate per-instance reporting alongside aggregate scores, and Henderson et al. (2018) and Agarwal et al. (2021) address seed sensitivity and uncertainty in reinforcement learning. The authors resample runs within tasks, with an additional task-resampling variant in their Appendix A.5, while our application requires preservation of paired run effects across benchmarks.

Hardy et al. (2026) apply generalizability theory to 4,000 released models, which places replication across models rather than training seeds. Kohli (2026) estimates roughly two independent votes among nine frontier judges, and Messing (2026) reports naive standard errors that are 40 to 60 percent too small after accounting for judge and prompt variation. Sha and Zhao (2026) estimate how many independent signals a leaderboard’s benchmark scores provide across models, which counts variation between different models rather than the run-to-run covariance at a fixed configuration that our effective count describes. Those studies measure dependence across models, judges, or prompts, while we measure paired benchmark deviations across training runs.

The cross-half identity builds on the repeated-measurement approach to attenuation described by Spearman (1904). Under independent errors between halves, paired measurements remove the item-noise contribution from covariance estimates, including diagonal entries. Cronbach (1951) relates split-half reliability to random halves of an item population, which motivates the reliability table and the 50 re-splits in N3. Those re-splits change the margin estimate by a reported standard deviation of 0.002.

The Brown (1910) and Spearman (1910) test-length formula contains $1 + (k-1)\rho$, the same algebraic form used for Λ^2 . Its correlation concerns parallel items, whereas ours concerns benchmark run effects. Substituting the reliability means in Table 2 at $k = 10$ would produce 6.87 and 3.91, which aren’t estimates of our variance inflation.

Campbell and Fiske (1959) compare traits and measurement methods in a crossed matrix. The within-format and cross-format summaries here resemble heterotrait blocks, but no benchmark appears in multiple formats. This design therefore can’t separate benchmark content from scoring format or establish format as the cause of the association. Brennan (2001) provides the variance-component framework for these distinctions.

Hofmann et al. (2025) fit benchmark-specific item-response models to an overlapping evaluation panel, with seven authors shared with Heineman et al. (2025). Kipnis et al. (2025) estimate a factor across six benchmarks that correlates 0.94 with total score. Those ability structures describe performance differences rather than the within-configuration noise covariance estimated here. Hedges et al. (2010) estimate uncertainty for dependent effect sizes without requiring their complete covariance matrix, while our application estimates the covariance needed for an effective-count point estimate.

Covariance diagnostics The largest discrepancy between opposite cross-half covariance entries is 2.44 bootstrap standard errors. This maximum across multiple pairs doesn’t by itself establish leakage. The reduced item records provide 57 MMLU subject groups but only one group for each other trait. The passage-aware split in Appendix A uses the release’s request files instead.

Among the eight accuracy traits with positive estimated seed variance, estimated correlations range from -0.216 to 1.296 , with the maximum occurring for ARC-Easy and MMLU. The symmetrised matrix has a negative eigenvalue of -0.55 and a leading eigenvalue of 3.85 . These estimates don't form a valid correlation matrix, a possible finite-sample outcome of noise-corrected moment estimation. We retain the unprojected matrix because a positive-semidefinite projection would change the reported values without an established correction target. The primary ratio uses cross-half sums directly and doesn't require this correlation-matrix transformation. The nine-trait margin matrix has correlations from -0.157 to 0.802 and nine positive eigenvalues. Its leading eigenvalue is 4.33 , with a spectral participation ratio of 3.57 . The accuracy participation ratio is 3.12 before clipping its negative eigenvalue and 3.62 after clipping. Because that accuracy matrix is indefinite, neither figure provides a validated count of independent population directions.

Table 5. Exploratory scorecard for the eleven reported planning predictions. Several outcomes depend on interpretations absent from the planning document, and the table records those limitations.

No.	Criterion	Measured result	Assessment
1	Margin above 1.349, accuracy above 1.40, lower limits above 1	1.244 and 1.078	Neither point threshold reached
2	Accuracy-to-margin ratio at least 0.8	0.867	Point criterion met
3	Retained excess above 0.70 and N5 share below 0.30	1.23 and 0.61	Default N5 criterion fails
4	N6 share below 0.50	0.015	Point criterion met
5	Underprediction in at least four bands	Intervals exclude 1 in one band	Fails interval reading, meets point reading
6	Transport within 20 percent, independence error above 20 percent	+66%, -28.9% , three units	Excess-scale criterion fails, raw scale differs
7	External estimate and lower limit above 1	1.705 for likelihood	Required interval unavailable
8	At least eight ratios in $[1.4, 3.0]$	Ten, range $[1.571, 1.732]$	Lower threshold can't discriminate
9	Cross-format diagnostic above 1.2, lower limit above 1	0.921 and 0.965	Neither criterion met
10	Frobenius distance above 0.5	5.59 and 6.49	Criterion met with fill convention
11	More than ten percent of pairs with predicted flip probability above 0.05	455 of 1,500, or 30.3 percent	Observed-gap calculation meets criterion

The original scorecard counted five failures, four passes, and two predictions that couldn't be tested as written. Predictions 9 and 11 came from a separate analysis of the per-run reductions, rather than the original pipeline. Prediction 7 lacks its required lower confidence limit, and prediction 6 depends on a post hoc choice to compare excess inflation. Table 5 retains the numerical outcomes and states where the assessment differs from the proposed criterion.

Variation across model sizes Each size band contains 25 configurations, one per recipe. Its clusters are therefore singletons, without within-band replication of sizes inside recipes. Only the 530M interval excludes one on either score scale, as Table 6 reports, and that margin exclusion disappears under the test-inversion interval of Appendix C. The intervals don't support a ranking of model sizes. All five margin point estimates and four accuracy point estimates exceed one, so prediction 5 passes a point-estimate interpretation and fails the stricter interval interpretation used in the original scorecard.

The lower squared-ratio bound for 1B margins falls below zero, which leaves its square-root endpoint unavailable. Three constructions that stay positive give the 1B margin estimate finite endpoints, and none of them excludes one. The configuration percentile bootstrap gives 0.834 to 1.643 and the cluster-robust $t(24)$ interval on the log of the squared ratio gives 0.811 to 1.628 , while a wild cluster bootstrap-t on that log scale gives 0.491 to 1.656 . At the full battery the log-scale wild

Table 6. Inflation estimates by model size across the 25 recipes. The intervals use the wild cluster bootstrap-t calculation described in Appendix D.

Size	Λ_M	95% interval	Λ_A	95% interval
150M	1.101	[0.781, 1.344]	1.028	[0.878, 1.165]
300M	1.350	[0.638, 1.803]	1.080	[0.836, 1.288]
530M	1.514	[1.064, 1.837]	1.219	[1.083, 1.338]
750M	1.110	[0.686, 1.407]	0.987	[0.310, 1.320]
1B	1.149	[n/a, 1.751]	1.205	[0.805, 1.497]
Pooled, $N = 125$	1.244	[1.143, 1.338]	1.078	[0.993, 1.157]

interval is 1.145 to 1.338, which agrees with the reported 1.143 to 1.338 at the second decimal. The transform matters only where the squared ratio is poorly determined.

That same run also estimated every size band with BoolQ removed, and those cells sit well away from one. On margins they give 1.548, 1.864, 1.874, 1.888 and 1.649 from 150M to 1B, with intervals of [1.325, 1.742], [1.763, 1.961], [1.694, 2.039], [1.660, 2.082] and [1.215, 2.001], so every band excludes one and the 1B cell recovers a finite lower endpoint that the full battery can't supply. On accuracy the same removal gives 1.842, 1.900, 1.548, 1.594 and 1.222, with intervals of [0.989, 2.408], [1.174, 2.417], [1.262, 1.792], [1.341, 1.811] and [0.429, 1.674]. The middle three exclude one while the smallest and the largest band don't. We read this as the composition sensitivity reappearing at size resolution rather than as independent support for the pooled estimate, because the full-battery sweep already names BoolQ as the benchmark pulling hardest toward one.

A separate run scored those constructions where they actually get used, inside one size band at the three runs the release ships, and the transform doesn't repair that case. On a margin-like band the plain interval covers 0.930 and the log one covers 0.902, and at ten runs the two sit at 0.947 and 0.944. On an accuracy-like band at three runs the plain interval is undefined in 0.284 of replicates and misses in 0.017 of them, while the log interval is undefined in 0.090 and misses in 0.151, so it buys definedness by paying coverage. The leverage correction is the better repair of the two, since it lifts the margin band to 0.942 at three runs and the accuracy band to 0.952 at ten. The 1B log-scale endpoints should therefore be read as closer to a ninety percent statement than a ninety five percent one, which widens the caveat that row already carries (seed 20260972, 4,000 replicates).

At the full 125-cell design the two constructions agree instead, covering within 0.006 of each other on an independent population, at the band-shared bound, at the recipe-shared bound, at the joint corner and even at a band-shared quarter where both fail together near 0.82. The leverage correction is the only construction that improves on either, holding 0.954 at the corner. The transform is a device for keeping endpoints positive in a thin cell rather than a better interval (seed 20260978, 4,000 replicates).

Simulation says these per-size rows should be read as estimates with wide uncertainty rather than as intervals carrying their nominal level. The accuracy column is the weaker of the two. One band of 25 configurations at the three runs the release ships covers 0.930 on a margin-like population and only 0.707 on an accuracy-like one, where the louder item noise leaves 25 clusters too little to work with. Six runs repair the accuracy case to 0.937 and ten reach 0.947. The shortfall is a sample size problem rather than a flaw in the interval. The power to detect anything inside a single band is worse again than the coverage. At a true margin ratio of 1.2446 a single band excludes one in 0.313 of replicates at three runs, 0.739 at six and 0.957 at ten, while at a true accuracy ratio of 1.078 the exclusion rate reaches only 0.081 at ten runs and 0.317 at forty. No affordable run count settles the accuracy question inside one band, which is the honest reason the paper draws its conclusions from the pooled design. These ran at 4,000 replicates with seeds 20260963 and 20260971.

Figure 1 reports N1 calibration at six effect sizes with 2,000 replicates each, using the 2.5th and 97.5th percentiles of estimates as dispersion bands, and Table 9 gives those recovered estimates alongside the other null and sensitivity runs. Those bands describe simulated estimator dispersion rather than confidence-interval coverage.

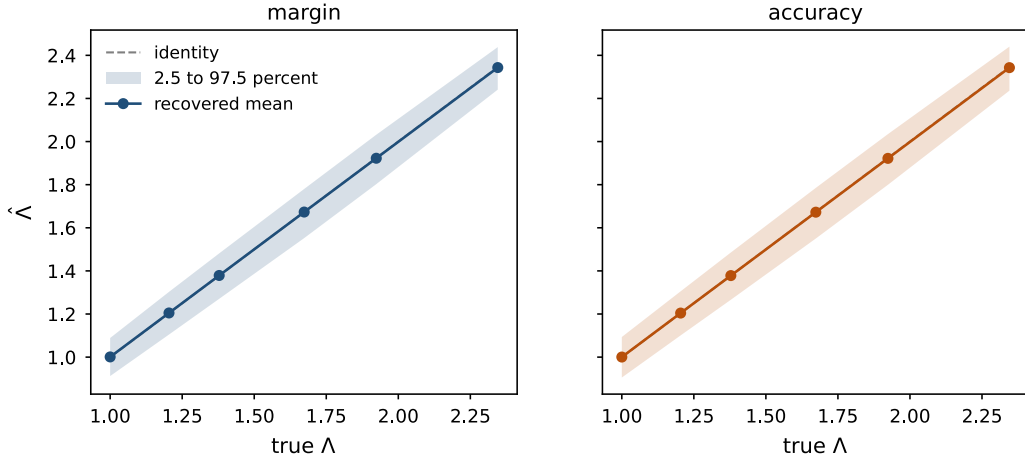


Figure 1. Reported simulation recovery across six specified covariance settings for both score scales. The shaded bands describe replicate dispersion and don’t establish interval coverage under other data-generating assumptions.

Scoring-scale interventions We transformed the saved margins item by item and recomputed each estimate with the items, runs, split, and interval routines held fixed. For benchmark j , s_j is the standard deviation of its per-item margins pooled across all configurations, runs, and items, and it ranges from 0.087 for WinoGrande to 0.468 for CommonsenseQA. Table 7 reports the untransformed margins, division by s_j , clipping at $\pm s_j$ before that division, two hyperbolic tangent scores, and the sign of each margin. Multiplying every margin by ten leaves the estimate unchanged to nine decimal places, which checks the construction.

Division by s_j changes only the benchmark weights. It lowers margin inflation from 1.244 to 1.182. The three bounded scores lower it further, to between 1.120 and 1.141. The sign score gives 1.079, which differs from the accuracy estimate of 1.078 by 0.0007 for reasons we haven’t traced. Across these scores the off-diagonal sum falls from 0.547 to 0.164 times the trace, while BoolQ’s share of that trace grows from 0.679 to 0.844. Bounding the margins of the same runs therefore reproduces between 62 and 75 percent of the gap between margin and accuracy inflation. These transformations change the scoring rule and leave the trained models untouched, so they don’t show how training produces large margins, and the intervals treat each s_j as fixed.

Table 7. Margin inflation after item-level transformations of the same saved margins, with wild recipe-cluster intervals over 4,999 draws. s_j is the pooled standard deviation of benchmark j ’s item margins.

Item score	$\hat{\Lambda}$	Interval	Off-diagonal sum / trace	BoolQ trace share
m	1.244	[1.143, 1.338]	0.547	0.679
$10m$	1.244	[1.143, 1.338]	0.547	0.679
m/s_j	1.182	[1.094, 1.265]	0.398	0.786
$\min(\max(m, -s_j), s_j)/s_j$	1.134	[1.048, 1.213]	0.287	0.827
$\tanh(m/s_j)$	1.141	[1.053, 1.222]	0.301	0.822
$\tanh(2m/s_j)$	1.120	[1.032, 1.200]	0.254	0.831
$\text{sign}(m)$	1.079	[0.992, 1.158]	0.164	0.844

Table 8 applies $\bar{r}_E = 0.0181$ while holding trait standard deviations fixed. Its adjusted column applies to the weighting of this battery, with flip probabilities interpreted under the stated Gaussian comparison model. The held-out accuracy point errors favour the full matrix over independence, 0.0105 versus 0.0143, without an interval for their difference. Every predictor also receives held-out seed standard deviations unavailable without replication. These results don’t establish an operational replacement for replicated runs. The three-unit PolyPythias accuracy estimate of 2.286 differs from

Table 8. Accuracy-point calculations at the median 300M configuration. The four- and seven-trait rows evaluate the same ten-trait quantities at different K values and aren't additional measured batteries.

K	Independent σ_{agg}	Adjusted σ_{agg}	K_{eff}	P_{flip} , one-point gap
4	0.754	0.774	3.79	0.181
7	0.570	0.600	6.31	0.119
10	0.477	0.514	8.60	0.085

1.078 and has a configuration-bootstrap interval with no defined lower endpoint, which further limits transport of this calculation.

Table 9. Reported null and sensitivity results with replicate standard deviations. N3 and N4 use the observed correlation and therefore describe sensitivity rather than a zero-correlation null.

Test	Structure	$\hat{\Lambda}_M$	SD	$\hat{\Lambda}_A$	SD
N1	Zero correlation with observed counts and reliabilities	1.0009	0.046	1.0004	0.048
N2	Independent trait-specific run-label permutations	1.0010	0.048	0.9984	0.039
N1	$\bar{r}_E = 0.05$, true $\Lambda = 1.2042$	1.2048	0.051	1.2044	0.053
N1	$\bar{r}_E = 0.10$, true $\Lambda = 1.3784$	1.3788	0.055	1.3782	0.057
N1	$\bar{r}_E = 0.20$, true $\Lambda = 1.6733$	1.6730	0.059	1.6724	0.061
N1	$\bar{r}_E = 0.30$, true $\Lambda = 1.9235$	1.9226	0.059	1.9220	0.061
N1	$\bar{r}_E = 0.50$, true $\Lambda = 2.3452$	2.3433	0.052	2.3429	0.053
N3	Fifty independent item re-splits	1.2459	0.002	1.0843	0.008
N4	Zero mediation with pooled-slope regression	1.0910	0.096	1.1045	0.062
N5	Zero correlation plus gain, $\sigma_g = 0.05$	1.1477	0.050	0.9986	0.048
N6	Zero correlation plus default-versus-auxiliary offset	1.0038	0.044	1.0030	0.047
N6	Same offset with auxiliary-run contrast	1.0004	0.062	1.0003	0.065

Paired uncertainty for the prediction comparison Table 10 gives the fold-mean prediction errors that Section 4.4 summarises for each of the six models. Before fitting any model we clip negative variance estimates, while the target retains the original covariance contributions.

Table 10. Mean squared prediction error of log aggregate standard deviation across five recipe folds, with paired recipe-bootstrap intervals for their differences given below. The rescaled plug-in is an exploratory addition to the original five-model comparison.

Model	Margin	Accuracy
Independence	0.0358	0.0143
Rank-one competence	0.0057	0.0140
Rank-one plus gain	0.0063	0.0260
Phenotypic correlation plug-in	0.1073	0.1682
Full covariance	0.0068	0.0105
Rescaled phenotypic plug-in	0.0131	0.0342

The rank-one model has the lowest margin point error, and full covariance has the lowest accuracy point error, but paired recipe-bootstrap intervals with 2,000 refitted draws separate neither model from its runner-up. On margins the rank-one, rank-one plus gain, and full models each lower the error relative to independence by about 0.03 with intervals that exclude zero, although the rank-one interval reaches zero when nonpositive held-out variances are filled from training folds. On accuracy every computed interval for a difference from independence includes zero, as the paired intervals below show. The unrescaled phenotypic plug-in has the largest error on both scales. Rescaling its off-diagonals on training folds reduces that error, though the rescaling needs replicate information and still trails the best model on either scale.

We reran the five-fold comparison with every prediction, target, and squared log error saved, and the rerun reproduces all twelve fold means in Table 10 exactly, with P2R factors of 0.366 and 0.370. Before computing any contrast, we fixed eight paired differences and a practically meaningful difference of 0.005 in mean squared log error, which corresponds to a 7 percent error in the aggregate standard deviation. For intervals, we resampled the 25 recipes 2,000 times, assigned folds over the distinct original recipes so that copies of one recipe never sit on both sides of a split, and refitted the whole comparison in each draw. These intervals treat the designed recipes as exchangeable clusters. They also absorb variation from the choice of fold partition.

An operational comparison instead predicts each held-out configuration’s raw aggregate variance moment from matrices refitted without held-out marginals. There, independence has mean squared error 5.01×10^{-9} on margins and 1.72×10^{-9} on accuracy. The full, rank-one, and shrinkage models differ from it by amounts whose 2,000-draw recipe-resampling intervals all include zero. That target is the noisy three-run moment, so the comparison bounds detectable gains without calibrating the ranking above.

The 1,500 same-size pairs of the flip census also have seed-matched runs. The paired difference carries a positive variance estimate for 1,445 margin and 1,222 accuracy pairs. For those pairs the ratio of the paired-difference standard deviation to the independence value has median 1.024 on margins and 1.081 on accuracy, with 5th to 95th percentiles of 0.39 to 1.40 and 0.36 to 1.62. The stricter subset whose two configurations both carry positive variance holds 1,379 and 919 pairs at medians of 1.027 and 1.079. In 0.49 of pairs the seed-matched covariance is positive.

On margins, P1, P1G, and P3 each lower the error relative to independence by about 0.03, with intervals of -0.058 to -0.001 , -0.059 to -0.002 , and -0.060 to -0.004 , and between 95 and 97 percent of draws put the reduction beyond 0.005. P2R lowers the error by 0.023, but its interval of -0.059 to 0.010 includes zero. The P1 minus P3 interval runs from -0.003 to 0.012 . These data don’t separate the rank-one model from the full matrix. On accuracy, each of the four intervals for a difference from independence includes zero. The full matrix minus independence runs from -0.017 to 0.023 .

Figure 2 draws all eight contrasts with both sources of uncertainty side by side. Across 1,000 random five-fold partitions of the original recipes, independence never has the lowest margin error, while P1, P1G, and P3 have the lowest error in 27, 33, and 33 percent of partitions. On accuracy, P3 has the lowest error in 55 percent of partitions and independence in 39 percent. The partition behind the table understates the margin advantage of the correlation models relative to a typical partition, since its independence error of 0.0358 lies below the partition median of 0.0438 and its P1 error of 0.0057 lies above the median of 0.0031. These contrasts concern correlation prediction with held-out marginal variances supplied, so they don’t carry over to the operational comparison in the main text, where no model separates from independence.

We also ran the comparison on simulated populations with known seed covariance, using 300 replicates for each of the three structures in Table 15, and scored every model against both the noisy fold target and the true aggregate standard deviation. The noisy target itself has a mean squared log error against the truth of 0.011 to 0.012 on margins and 0.024 to 0.032 on accuracy. With the matched margin covariance, P1, P1G, and P3 beat independence by 0.043 to 0.046 against the noisy target and by 0.048 to 0.051 against the truth, and P3 has the lowest true error, 0.0027 below P1. With independent seed deviations, independence has the lowest margin error under both targets, and P1, P1G, and P3 lose between 0.001 and 0.007. On accuracy with the matched covariance, the true errors of P0, P1, P1G, and P3 differ by at most 0.0032, which is small beside the target noise. A five-fold accuracy comparison shouldn’t be expected to separate them. These populations share one covariance across recipes and contain little systematic variation between configurations, so they don’t reproduce how the performance-correlation plug-ins behave. We don’t use them to assess P2 or P2R.

Recipe, size, and trait sensitivity Removing each recipe in turn gives margin estimates from 1.207 to 1.256 and accuracy estimates from 1.065 to 1.099. The maximum changes from the complete battery are 0.037 and 0.021. Removing each size gives ranges of 1.178 to 1.289 and 1.047 to 1.115, compared with individual-band ranges of 1.101 to 1.514 and 0.987 to 1.219. These overlapping intervals don’t establish a size trend in the effect.

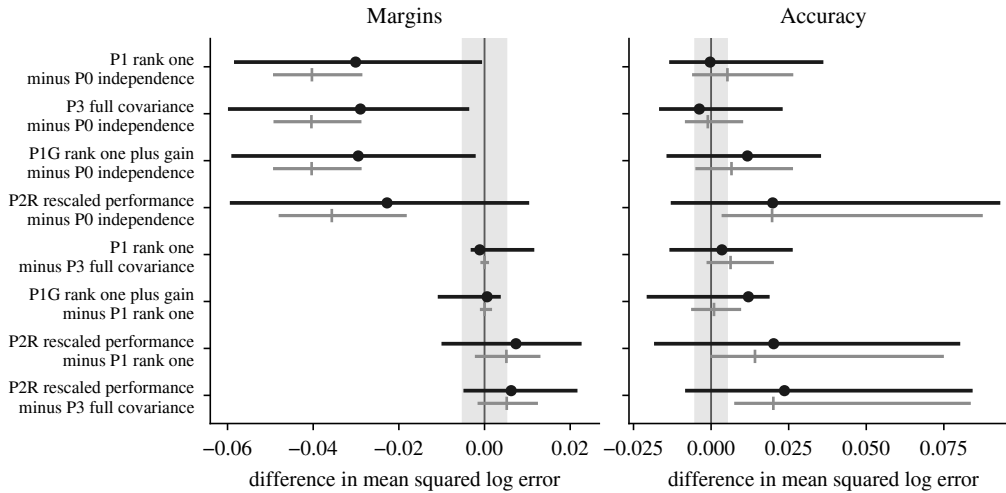


Figure 2. Paired prediction differences for the eight contrasts fixed before any of them was computed, in mean squared error of the log aggregate seed standard deviation. Black marks the recipe bootstrap, resampling all 25 recipes over 2,000 refitted draws, with the observed point estimate as the dot. Grey marks 1,000 fold repartitions of the same 25 recipes, with the partition median as the tick, so the two rows separate recipe sampling from fold assignment. The shaded band is the practically meaningful difference of 0.005 that we fixed beforehand, and a negative value favours the first model of the pair. Every interval is a 2.5 to 97.5 percentile range rather than a standard error, and no endpoint here is missing or clipped.

Trait removal changes the estimates more than recipe or size removal. Excluding BoolQ gives 1.786 with interval 1.689 to 1.877 for margins, and 1.558 with interval 1.412 to 1.687 for accuracy. Excluding CommonsenseQA gives the smallest point estimates, 1.096 and 1.014. BoolQ contributes 68 percent of the estimated margin variance trace and 84 percent of the accuracy trace. The removal estimates imply that the off-diagonal terms involving BoolQ sum to about -0.15 of the margin trace and -0.06 of the accuracy trace. Its removal therefore lowers the denominator by more than the numerator. The same removal lowers the aggregate standard deviation from 5.72×10^{-3} to 5.16×10^{-3} on margins and from 5.14×10^{-3} to 3.26×10^{-3} on accuracy. A larger inflation factor here accompanies a less variable average. Removing WinoGrande is the only other removal whose accuracy interval, 1.023 to 1.178, excludes one, and Table 11 gives the remaining intervals and standard deviations. Removal is a switch, so we also turned it into a dial, lowering BoolQ’s weight below the flat tenth in seven steps to 0.0533 while raising the other nine weights together to keep the battery summing to one. Margin inflation climbs from 1.2439 at the flat weighting through 1.3374 at eight percent to 1.5030 at the smallest weight we tried, and every interval along the way excludes one. Accuracy climbs from 1.0784 to 1.2311 over the same steps. Its interval stops including one at a BoolQ weight of 0.0867, which is a reduction of about one seventh. The accuracy conclusion is therefore a knife edge in the weighting rather than in the estimate, which moves smoothly, since a battery that discounts BoolQ by a seventh reports dependence on accuracy as well. We report the flat weighting because the release gives no basis for singling out BoolQ. A reader who has such a basis should expect the accuracy answer to move first.

The variance-share participation ratios are 1.97 for margins and 1.42 for accuracy on traits with defined seed variance. The first two ratios measure how strongly a few traits dominate variance, while the noise-correlation participation ratio of 3.57 measures concentration across matrix directions.

Standardising traits by estimated seed standard deviation gives 1.803 on nine margin traits and 1.707 on eight accuracy traits. These point diagnostics omit traits with nonpositive variance estimates and divide by uncertain standard deviations, with seven accuracy traits at or below item noise. Holding those estimated weights fixed, the wild recipe-cluster intervals are 1.544 to 2.023 and 1.228 to 2.076. These intervals omit the uncertainty in the weights themselves. We don’t score them against the planning thresholds.

Table 11. Trait-removal estimates across all 125 configurations on both score scales, with wild recipe-cluster intervals over 4,999 draws. Each removal row uses the nine remaining traits with equal weights of one ninth, and the final row uses all ten traits. The columns σ_{agg} and σ_{ind} give the aggregate and independence standard deviations in units of 10^{-3} , so the weighted trace equals σ_{ind}^2 and the weighted off-diagonal sum equals $\sigma_{\text{agg}}^2 - \sigma_{\text{ind}}^2$.

Removed trait	Λ_M	Interval	σ_{agg}	σ_{ind}	Λ_A	Interval	σ_{agg}	σ_{ind}
BoolQ	1.786	[1.689, 1.877]	5.16	2.89	1.558	[1.412, 1.687]	3.26	2.09
WinoGrande	1.246	[1.144, 1.340]	6.36	5.11	1.102	[1.023, 1.178]	5.87	5.32
HellaSwag	1.244	[1.142, 1.338]	6.35	5.10	1.076	[0.992, 1.153]	5.69	5.29
PIQA	1.243	[1.139, 1.340]	6.35	5.10	1.073	[0.988, 1.150]	5.69	5.30
MMLU	1.205	[1.103, 1.300]	6.13	5.09	1.068	[0.984, 1.143]	5.66	5.30
ARC-Challenge	1.195	[1.113, 1.272]	6.08	5.09	1.039	[0.952, 1.116]	5.47	5.27
OpenBookQA	1.178	[1.092, 1.257]	5.97	5.07	1.074	[0.992, 1.150]	5.66	5.27
ARC-Easy	1.161	[1.089, 1.228]	5.81	5.01	1.046	[0.957, 1.126]	5.43	5.19
Social IQa	1.139	[1.068, 1.205]	5.68	4.99	1.032	[0.950, 1.109]	5.41	5.24
CommonsenseQA	1.096	[1.054, 1.137]	4.99	4.55	1.014	[0.936, 1.086]	5.15	5.08
None	1.244	[1.143, 1.338]	5.72	4.59	1.078	[0.993, 1.157]	5.14	4.77

Removing defined traits from the standardised batteries leaves margin estimates between 1.625 and 1.910 and accuracy estimates between 1.388 and 1.780, with BoolQ removal at 1.910 and 1.761 under this standardisation. Selecting traits by positive estimated variance can change the target. It could raise the mean correlation if the excluded low-signal traits also have low true correlations. We haven't tested that additional assumption, and subset stability doesn't resolve uncertain denominators.

The raw trait-removal spans are 0.690 for margins and 0.544 for accuracy, compared with interval widths of 0.195 and 0.164. The resulting ratios of 3.5 and 3.3 compare different measures of spread and aren't uncertainty decompositions. Applying the delete-one-trait jackknife formula produces scales of 0.554 and 0.455, compared with sampling standard errors of 0.045 and 0.038. Their ratios are 12.3 and 12.1, partly because the jackknife multiplier is 2.85. Comparing the unscaled dispersion instead gives ratios of 4.33 and 4.24.

Omitting the BoolQ removal value from that calculation gives scales of 0.139 and 0.073, or 3.08 and 1.93 times the reported sampling standard errors. These are descriptive composition diagnostics, because the ten fixed benchmarks aren't established exchangeable draws from a benchmark population. Adding the original jackknife scales and sampling standard errors in quadrature gives 0.556 and 0.457. A normal calculation then gives ranges of 0.15 to 2.33 and 0.18 to 1.974, both including one. Repeating that arithmetic without the BoolQ removal value gives 0.958 to 1.529 and 0.918 to 1.239.

Those calculations aren't validated confidence intervals for a benchmark population, since they can double-count sampling variation and their normal approximation behaves poorly near the parameter boundary. For example, $\Lambda = 0.15$ corresponds to $\bar{r}_E = -0.109$, only approximately 0.003 above $-1/9$. We retain the fixed-battery target and report composition sensitivity without treating these scales as an identified decomposition of total uncertainty.

Changing trait count also changes Λ at fixed effective covariance coefficient. At $\bar{r}_E = 0.061$, the nine-trait reference is 1.219 rather than the ten-trait 1.244, while the corresponding accuracy values are 1.070 and 1.078. Four margin removals and five accuracy removals exceed those mechanical references. On eight traits, standardised accuracy inflation of 1.707 implies 2.75 effective benchmarks, while the raw accuracy effective coefficient of 0.0181 implies 7.10 effective benchmarks. Comparing 2.75 with 7.10 holds trait count fixed, unlike comparison with the ten-trait value of 8.60.

The ARC pair carries an estimated margin correlation of 0.801, and removing both of its benchmarks leaves margin inflation at 1.118 with interval 1.059 to 1.174 and accuracy inflation at 1.012 with interval 0.929 to 1.086. The pair affects the estimate, although the remaining margin interval still excludes one.

Benchmark count, identity, and weights To separate benchmark count from benchmark identity, we computed both estimates for all 1,013 equal-weight subsets of two to ten benchmarks, each with a wild recipe-cluster interval over 4,999 draws. Table 12 summarises these subsets separately for each size K . Median margin inflation lies between 1.012 and 1.244 across sizes. The share of subsets whose interval lies above one rises from 38 percent at two benchmarks to 98 percent at eight. Accuracy medians stay between 1.017 and 1.078, and at each size from two to nine benchmarks between 20 and 31 percent of accuracy intervals lie above one. Among subsets of two to five benchmarks, 16 accuracy subsets have a nonpositive independence variance or a negative estimate of squared inflation, and so no defined estimate, which the table’s quantiles omit. Subsets move one weight to zero and leave the rest flat, which probes only the corners of the weighting. We also rebuilt the statistic under arbitrary weights and swept the interior. Rebuilding the numerator and the denominator from the same contrast projections reproduces the shipped values to 1e-19 under flat weights, which is the check that licenses the sweep. Over 2,000 weightings drawn from a flat Dirichlet, 0.817 of margin intervals still exclude one, and the estimate runs from 0.976 to 1.991 with a 5th percentile of 0.999. Over 2,000 weightings drawn from a Dirichlet concentrated near the flat point, every margin interval excludes one and the lowest estimate is 1.059. Accuracy behaves in the direction that favours the reported claim, because 0.386 of flat-Dirichlet intervals and 0.402 of near-flat ones exclude one while the flat weighting itself gives 0.993 to 1.157 and doesn’t. The weighting we shipped is therefore the conservative one on accuracy. The conclusion on margins holds across four fifths of the simplex and all of its neighbourhood.

Holding the number of benchmarks fixed, subsets that contain BoolQ have mean margin inflation lower by 0.115 at two benchmarks and by 0.596 at nine. The corresponding accuracy differences are 0.076 and 0.499. At nine benchmarks every other benchmark raises mean margin inflation, by amounts from 0.004 for WinoGrande to 0.171 for CommonsenseQA. Pooled across sizes, margin estimates have a median of 1.136 with quartiles 1.018 and 1.390, and they run from 0.927 for HellaSwag with WinoGrande up to 1.786 in subsets that drop BoolQ and WinoGrande. Accuracy estimates have a median of 1.038 with quartiles 1.001 and 1.229, and they run from 0.315 to 5.439. Two thirds of margin intervals exclude one, always from above, against 0.305 of accuracy intervals. A separate implementation recomputed every subset and reproduced each size’s median and share, together with the full-battery 1.244 with 1.143 to 1.338 and 1.078 with 0.993 to 1.157. The subsets share runs, items, and configurations, so their spread describes this battery and doesn’t sample a wider population of batteries.

Weighting by item count changes the weights of all ten benchmarks without selecting on scores. It gives margin inflation of 1.119 with interval 1.068 to 1.170 and accuracy inflation of 1.043 with interval 0.963 to 1.119, while MMLU and HellaSwag carry 64 percent of the weight. A second reweighting, by the inverse pooled seed standard deviation, reproduces the standardised diagnostic above, which has to drop the benchmarks with nonpositive variance estimates.

Table 12. Equal-weight subsets of the ten benchmarks, grouped by size K . Quantiles cover subsets with a defined estimate, and the share above one counts subsets whose wild recipe-cluster interval lies entirely above one.

K	Subsets	Margin				Accuracy			
		Median	10th pct.	90th pct.	Above one	Median	10th pct.	90th pct.	Above one
2	45	1.012	0.988	1.267	38%	1.017	0.918	1.226	24%
3	120	1.137	0.987	1.368	54%	1.020	0.914	1.305	29%
4	210	1.210	0.986	1.467	59%	1.030	0.958	1.344	29%
5	252	1.138	0.990	1.562	62%	1.035	0.973	1.398	31%
6	210	1.103	1.009	1.607	70%	1.041	0.979	1.437	30%
7	120	1.131	1.034	1.675	86%	1.042	0.988	1.467	31%
8	45	1.160	1.066	1.692	98%	1.046	1.007	1.497	29%
9	10	1.200	1.135	1.300	100%	1.071	1.031	1.148	20%
10	1	1.244	1.244	1.244	100%	1.078	1.078	1.078	0%

Observed gaps and predicted rank reversals Using accuracy noise $\sigma_{\text{agg}} = 0.00514$, the Gaussian calculation gives $P_{\text{flip}} > 0.05$ for a true gap below 0.01196. Among the $5\binom{25}{2} = 1,500$

observed same-size pairs, 455 have observed gaps below this threshold, or 30.3 percent. The proportions range from 28.7 to 33.3 percent across bands. Substituting observed gaps meets the stated numerical criterion, but noisy observed gaps can move pairs across the threshold in either direction. This count therefore isn't a lower bound on the fraction of true gaps below the threshold, and it doesn't directly measure realised reversal frequency.

A practitioner faces the operational form of the same question, which is whether correcting their own test for this covariance helps once the correction carries the noise of a three-run design. We simulated that question directly rather than arguing it. Each of 4,000 replicates builds the shipped design, estimates the ratio from it, and then applies three rules to a fresh comparison of two model averages. The three are an uncorrected rule that assumes benchmark independence, a plug-in rule that multiplies its standard error by the estimate, and an oracle rule that uses the true value. On a margin-like population the uncorrected rule declares a difference in 0.113 of replicates when the true gap is zero, against its nominal 0.05, while the plug-in rule declares in 0.051 and the oracle in 0.050. Three runs per model leave those at 0.115, 0.053 and 0.051. The estimate carries a standard deviation of 0.059 across replicates and still sits within 0.005 of the oracle in every cell. Its own noise costs almost nothing at this design. On an accuracy-like population the gap is smaller and the estimate noisier, at a standard deviation of 0.099, and there the three rules give 0.067, 0.052 and 0.047. The plug-in rule overshoots the oracle by 0.005 there, which is the price of estimating a correction that is itself close to one.

A practitioner who accepts that correction will ask next how many runs their own comparison needs, and whether buying runs also repairs the miscalibration. We swept the same three rules over one to ten runs per model at true gaps of zero, a quarter, a half and one aggregate standard deviation, with 4,000 replicates per combination. Extra runs buy nothing against the size error, because the true standard error and the independence standard error both fall with the square root of the run count and their ratio doesn't move at all. The sweep says so directly, since the uncorrected rule declares at a zero margin gap in between 0.112 and 0.122 of replicates across all ten run counts, while the corrected rule stays between 0.047 and 0.056, and the accuracy population gives 0.061 to 0.078 against 0.047 to 0.063. Neither sequence trends with the run budget, and each individual rate carries a Monte Carlo standard error near 0.004, so what looks like movement is the simulation rather than the design. Power is the thing that additional runs actually buy here. At a gap of one aggregate standard deviation the corrected rule declares in 0.100 of replicates with one run per model and in 0.614 with ten, while the accuracy population gives 0.111 and 0.615 at the same two budgets. No run count up to ten reaches four fifths at any of the three nonzero gaps. A calibrated comparison at this battery costs more runs than the release supplies. The correction fixes the error rate rather than the sample size.

A second sweep carried the same design out to 80 runs per model and found the size error exactly where it had been. Across 12, 16, 20, 24, 32, 40, 60 and 80 runs the uncorrected rule declares at a zero margin gap in between 0.108 and 0.119 of replicates, while the corrected rule holds between 0.047 and 0.053. Four fifths arrives at 16 runs per model for a margin gap of one aggregate standard deviation, which is exactly what the ordinary normal calculation for that gap gives, and at 20 runs on the accuracy population, where the oracle rule also misses four fifths at 16 runs by 0.002. The extra accuracy runs therefore pay for the normal approximation rather than for the estimated correction. A gap of half a standard deviation needs 63 runs per model by the same calculation. The nearest count above it that we simulated, 80, reaches 0.881 on margins. A third sweep took the quarter standard deviation gap out to 360 runs per model and found four fifths at exactly the 252 runs the closed form predicts for margins, and at 300 on the accuracy population, where even the oracle rule falls a fraction short at 252. The uncorrected rule still declares at a zero margin gap in between 0.112 and 0.122 of replicates at every one of those seven counts. The size error survives every run budget from one run per model to 360. Those are the numbers a replicate study should be sized against, and all of them sit far above the three runs per configuration that this release ships.

Both sweeps estimate the correction on the same population that the comparison is drawn from, which is the friendly case and not the one a reader is in. Anyone applying this correction will estimate the ratio once, on one battery and one model family, and will then carry it to a comparison whose own dependence they haven't measured, so we measured that transfer directly. The correction was estimated from the shipped margin design, whose true ratio is 1.245. It was applied to comparisons whose own true ratio was swept from 1.00 to 1.60, with 8,000 replicates per target and a matched rule re-estimated on the target for comparison. The transferred rule holds nominal size

only where the two ratios agree, at 0.053 when the target is 1.245 itself. Below that it turns conservative rather than wrong, declaring in 0.015 of replicates at a target of 1.00 and 0.028 at 1.10. It pays for that in power, since a gap of one aggregate standard deviation at a target of 1.00 is declared in 0.043 of replicates against 0.114 for the matched rule. Above it the rule under-corrects and the size error comes back, at 0.080 for a target of 1.40, 0.101 at 1.50 and 0.123 at 1.60, against 0.159, 0.188 and 0.216 for no correction at all. The matched rule holds between 0.048 and 0.053 across the entire sweep. The correction is therefore not a constant that a reader can borrow from this paper.

What transfers is the procedure rather than the number. A practitioner whose battery is more dependent than ours would keep most of the error they were trying to remove. Repeating the whole sweep with the accuracy design as the source, whose true ratio is 1.078 and whose estimate carries a standard deviation of 0.098, gives the same shape from the other end. That correction holds nominal size at 0.056 for a target of 1.10, which is its own neighbourhood. It recovers almost nothing above that, at 0.168 for a target of 1.50 against 0.199 with no correction at all and 0.054 for a matched rule. The penalty is therefore governed by the distance between the two ratios rather than by which design produced the number. Reading both sweeps together, the transferred rule keeps its declare rate between 0.04 and 0.06 only while the target ratio sits within about 0.05 of the source. That window is narrow enough that we recommend measuring rather than borrowing.

Adjacent-checkpoint variation in DataDecide We reduced every DataDecide run a second time at the largest checkpoint step that all three runs in its configuration share below the selected step. The gap is 1,250 steps at 530M and 750M, between 657 and 1,250 steps at 150M and 300M, and 1,869 steps at 1B. In two configurations, dclm-baseline-75p-dolma1.7-25p at 750M and dolma1.7-no-math-no-code at 1B, the three runs share no released step below the selected one, so the comparison covers 369 runs in 123 configurations. Table 13 reports both estimates at the two steps with the same items, split, weights, and interval routines.

Margin inflation falls from 1.233 to 1.192 at the earlier step. A paired recipe bootstrap over 4,999 draws gives a 95 percent interval of -0.078 to 0.001 for the change. Accuracy inflation rises from 1.082 to 1.130, with a change interval of -0.022 to 0.130 . Neither change interval excludes zero, but the accuracy interval at the earlier step, $[1.044, 1.210]$, excludes one. The reading of the full-battery accuracy estimate as close to independence depends on the scored checkpoint. A rerun with the seed-noise estimator on the same 123 configurations reproduces all four estimates. The centred test-inversion interval of Appendix C gives 1.059 to 1.222 for accuracy at the earlier step, so that exclusion of one doesn't depend on the wild construction. The same inversion gives 0.996 to 1.169 for accuracy at the selected step and lowers both margin lower endpoints, to 1.099 at the selected step and 1.080 at the earlier one. BoolQ's share of the accuracy trace moves only from 0.847 to 0.872 between the two steps.

In the form used for PolyPythias, the root-mean-square change in a run's aggregate score equals 0.469 between-seed standard deviations on margins and 0.631 on accuracy. After we remove each configuration's mean change, the ratios are 0.493 and 0.618. The movement differs between seeds and doesn't reflect shared training progress alone. Within configurations, the two steps' seed deviations correlate 0.870 for margins and 0.811 for accuracy. The three runs keep their order in 64 and 52 percent of configurations. The 750M band, where the selected step lies furthest from the final step, gives ratios of 0.889 and 1.048 with correlations of 0.502 and 0.511. The 750M margin ratio sits within 0.001 of the PolyPythias value of 0.890, while the accuracy ratio stays below the PolyPythias value of 1.573. The DataDecide gap is 1,250 steps rather than 1,000. The other four sizes give ratios between 0.220 and 0.551. We added this comparison after seeing the full-sample estimates. It is exploratory like the rest of the empirical study.

Table 13. Inflation at the selected step and at the previous step shared by all three runs, on the 123 configurations that have both. Estimate intervals use the wild cluster bootstrap-t, and the change interval uses a paired recipe bootstrap, both over 4,999 draws. The shift ratio is the root-mean-square change in a run's aggregate score divided by the between-seed standard deviation.

Scale	Selected step	Previous step	Change	Shift ratio	At 750M
Margin	1.233 [1.139, 1.319]	1.192 [1.104, 1.275]	$-0.041 [-0.078, 0.001]$	0.469	0.889
Accuracy	1.082 [0.992, 1.163]	1.130 [1.044, 1.210]	$0.048 [-0.022, 0.130]$	0.631	1.048

Simulation and permutation calibration The linearised margin standard error of 0.045 is close to the N1 null dispersion of 0.046, while accuracy gives 0.038 against the simulated 0.048, a difference of approximately 21 percent. The reported interval half-widths are 0.0974 and 0.082, approximately 2.2 standard errors compared with the $t(24)$ critical value of 2.06. Substituting the accuracy simulation dispersion gives a descriptive interval of 0.979 to 1.178, which also includes one. This substitution isn't a calibrated alternative interval because the simulated and observed populations differ.

A separate implementation of the interval definitions uses 25 clusters of five configurations, three runs, and ten traits. It assumes exchangeable seed covariance, independent item noise across halves, and recipe heterogeneity in seed variance. Across sixteen cells formed from four true correlations between zero and 0.10 and four noise settings, 200 datasets per cell give empirical coverage from 0.920 to 0.975. Each cell lies within two Monte Carlo standard errors of 0.95, and estimated inflation is within 0.01 of truth in fifteen cells. Four cells with dispersion 0.044, compared with observed null dispersions 0.046 and 0.048, have coverage 0.950, 0.935, 0.975, and 0.960. Their interval widths range from 0.203 to 0.238, compared with the reported margin width of 0.195. This limited simulation checks the procedure under its specified populations and doesn't validate all departures present in the study.

A second simulation, run for this revision with 2,000 replicates per population and 4,999 draws, scores four interval procedures on the same simulated moments. Table 14 lists the populations, each with 125 configurations in 25 recipe clusters, three runs, and ten benchmarks, and the wild bootstrap- t covers between 0.928 and 0.954. The cluster-robust $t(24)$ tracks it within 0.02 and falls to 0.911 with one recipe at six times the variance. The configuration percentile bootstrap falls to 0.872 under recipe-shared run effects. The recipe percentile bootstrap covers between 0.857 and 0.940 with undefined draws in the accuracy-like noise cells. Cross-half error correlation of 0.1 biases the estimate by -0.014 and -0.016 . Its margin-like wild interval misses above in 0.066 of replicates. The Monte Carlo standard error of a coverage near 0.95 is 0.005.

We later added a cluster test-inversion interval, a family the analysis plan lists for ratios with an uncertain denominator. For each candidate $\theta = \Lambda^2$, it forms the recipe sums $z_g = \sum_{c \in g} (T_c - \theta U_c)$ and keeps θ when the cluster-robust t statistic of their mean, with a $G/(G-1)$ factor, stays within the $t(24)$ critical value. The accepted set can be unbounded, and we record its shape instead of truncating it. Rerunning the R6 generator with the same seeds reproduces the cluster- t coverage of every row in Table 14. The inversion column scores the same 24,000 simulated datasets. The inversion covers between 0.932 and 0.957, with median widths within 0.006 of the wild interval. It holds 0.955 with one recipe at six times the variance, where the wild interval gives 0.930. Its set is bounded in 23,999 of those datasets, the one exception being a Student- t replicate. A restricted variant that imposes θ in the variance without centring covers between 0.946 and 0.970, which is above 0.95 in eleven of the twelve populations and more than two Monte Carlo standard errors above it in eight.

The Monte Carlo standard error of 0.005 at 2,000 replicates is the size of the shortfalls that Table 14 shows, so we repeated four cells with 10,000 replicates and new seeds. They are the two cells calibrated to the observed estimates and the two with the lowest wild coverage. The wild interval covers 0.933 with cross-half correlation 0.1, 0.935 with one dominant recipe, 0.940 in the margin-like cell, and 0.945 in the accuracy-like cell, each with a Monte Carlo standard error near 0.0025. We later repeated these four cells a second time at 10,000 replicates with a third seed, which returns 0.937, 0.935, 0.951 and 0.947. Three of them agree with the first run inside one standard error. The margin-like cell doesn't, because 0.940 and 0.951 sit more than three standard errors of their difference apart. We ran that one cell a third time at 40,000 replicates, which returns 0.947 with a Monte Carlo standard error of 0.0011 and puts the residual-scaled version at 0.955. Pooling all three runs of that cell over 60,000 replicates gives 0.946. It is the mildest of the four rather than a settled 0.940. The cross-half and dominant-recipe cells, which carry the shortfall that matters here, repeat within 0.004. In the cross-half cell the wild interval misses above in 0.054 of replicates against a nominal 0.025, so the shortfall falls on one side.

The centred inversion covers 0.935, 0.956, 0.942, and 0.947 in the same four cells. The inversion keeps its advantage of 0.021 under one dominant recipe while matching the wild interval elsewhere. We then repeated the other eight cells at the same precision, which puts every cell of Table 14 on

10,000 replicates for these four procedures. No cell moves by more than 0.011 against its 2,000-replicate value.

Across the twelve, the wild interval covers 0.933 to 0.953, the cluster-robust t covers 0.916 to 0.949, the centred inversion covers 0.935 to 0.956, and the restricted inversion covers 0.951 to 0.970. The restricted inversion covers at least as much as the centred one, which covers at least as much as the wild interval, which covers at least as much as the cluster-robust t , in eleven of the twelve cells. The exception is the cell with mean correlation 0.2, where the wild interval beats the centred inversion by 0.002. The three lowest wild coverages are 0.933 and 0.936 in the two cross-half correlation cells and 0.935 with one dominant recipe. A later sweep repeated all twelve populations once more at 10,000 replicates with a third seed, alongside the residual-scaled construction that the next paragraph compares, and it puts the wild interval between 0.929 and 0.950. That second sweep agrees with this one to within 0.004 at each end. The shortfall we describe here is a property of the construction rather than of one set of draws.

Coverage in simulation says which construction a reader should trust. It doesn't say how far the reported interval would move under a different one, so we applied all eight to the data itself. Each wild construction ran 9,999 draws over twenty bootstrap seeds. No endpoint varies across those seeds by a standard deviation above 0.0023. That matters most for the accuracy conclusion, whose lower endpoint of 0.9933 is the one number that decides whether the full-battery interval includes one. Its standard deviation across the twenty seeds is 0.0015. The endpoint sits more than four of its own bootstrap standard deviations below one. The inclusion of one is a property of the data rather than of the draws. On the full battery the margin interval runs from 1.141 to 1.339 under Rademacher weights, 1.166 to 1.344 under Mammen weights, 1.142 to 1.338 under Webb weights, and 1.130 to 1.349 under leave-one-cluster-out residual scaling. The cluster-robust t gives 1.147 to 1.334, the centred inversion gives 1.095 to 1.323, and the restricted inversion gives 1.074 to 1.329. Every one of the eight excludes one on that scale. On accuracy every one of the eight includes one instead. The widest of them, the restricted inversion, runs from 0.988 to 1.169. Removing BoolQ leaves both scales above one under all eight, where the widest margin interval is again the restricted inversion at 1.391 to 1.840. The shipped interval, which draws 4,999 times rather than 9,999, sits within 0.003 of the Rademacher mean in all four batteries. The draw count doesn't reach coverage either, since scoring 999, 2,499, 4,999 and 19,999 draws on identical replicates leaves coverage at 0.945 to 0.946 on an independent population and 0.941 to 0.943 at the band-shared bound, with median widths inside 0.002 of each other. A replicator on a laptop can therefore use the cheapest of those counts without changing a conclusion (seed 20260980, 4,000 replicates).

Every coverage number we report is a 95 percent number. A further run asks whether the procedure is calibrated or merely tuned at that one level. Scoring the same replicates at one in ten, one in twenty and one in a hundred gives 0.896, 0.951 and 0.990 on an independent population, and 0.896, 0.947 and 0.986 at the band-shared bound. The interval tracks the level it claims across the tails. The violation breaks all three levels together rather than one of them, since a band-shared quarter gives 0.743, 0.828 and 0.928, which is the same shortfall the main text reports at 95 percent (seed 20260979, 4,000 replicates). Which of these intervals include one, and which exclude it, doesn't depend on the construction we report.

None of that repairs the wild interval's own shortfall, so we asked whether a different construction would. We compared six of them on the same twelve populations with 4,000 replicates each. They are the wild bootstrap- t under Rademacher, Mammen and Webb weights, a wild bootstrap- t whose recipe residual is divided by $1 - h_g$ with h_g the recipe's share of the denominator sum, the cluster-robust t with the $G/(G - 1)$ correction, and the centred inversion. We ran four cells first, which are the two with the lowest wild coverage and the two calibrated to the observed estimates, and there the residual-scaled version reached nominal coverage in three of them while no other construction reached it in more than one. We then ran the remaining eight to see whether the advantage survived, and it did. Across all twelve at that replicate count it covers between 0.943 and 0.958, against 0.932 to 0.951 for the shipped Rademacher version. We then repeated both of those two constructions on all twelve populations at 10,000 replicates, which cuts the Monte Carlo standard error to 0.002 and puts the comparison at one precision on both sides. The residual-scaled version covers between 0.940 and 0.959 there, against 0.929 to 0.950 for the shipped one. It improves every cell by between 0.007 and 0.014. It reaches 0.95 in ten cells where the shipped version reaches it in one. Its intervals run 4 to 13 percent wider at the median. Mammen weights cover between 0.916 and 0.942 and lose to Rademacher in every cell, Webb weights land within 0.003 of Rademacher everywhere. The

corrected cluster-robust t covers between 0.923 and 0.950. The residual-scaled version also matches or beats the centred inversion in all twelve, by up to 0.012. We picked that construction after seeing the first four cells. The eight that followed confirm it on the same generator family rather than test it independently. We therefore report the shipped interval throughout and record this one as the better-calibrated alternative. On the observed data it gives [1.130, 1.349] for margins and [0.987, 1.163] for accuracy, so it moves no conclusion, except that its upper endpoint erases the 0.011 of separation from the planned threshold that we report below.

A reader can also ask what the reported interval becomes once its own miscalibration is corrected on these data, rather than through a construction chosen in simulation. We applied Beran pre pivoting, which runs a second wild bootstrap of 1,999 draws inside each of 1,999 outer resamples, and reads the outer quantiles at the levels the inner distribution implies. On the full battery the margin interval calibrates to [1.151, 1.328], which runs 0.018 narrower than the shipped [1.143, 1.338] and still excludes one. Accuracy calibrates to [0.994, 1.161] and still includes one, the BoolQ-removed margin interval calibrates to [1.689, 1.878], and the BoolQ-removed accuracy interval widens to [1.396, 1.704] while staying above one. The implied coverage of the nominal levels runs from 0.925 to 0.977 across the four batteries. This correction moves no conclusion, and on the full battery it disagrees with the simulation about the direction of the error. Pre pivoting measures calibration inside the resampling model that the interval already assumes. A violation the bootstrap shares with the estimator stays invisible to it. The simulated coverage remains the figure a reader should weigh. We tested that explanation rather than leaving it as an argument. Running the same pre pivoting inside the four simulated cells that carry the shortfall, with 1,000 replicates and 399 draws at each level, gives coverage of 0.924, 0.933, 0.931 and 0.944, against 0.938, 0.931, 0.936 and 0.953 for the shipped interval on the same replicates. An uncalibrated interval at the same 399 draws covers 0.936, 0.928, 0.938 and 0.949, so the draw count isn't doing the work. The calibration tracks the interval it corrects and repairs none of the shortfall, which is what a correction blind to the violation would do.

The estimator assumes that item noise in one half doesn't correlate with item noise in the other, so we asked how much a violation could manufacture. Holding the true seed correlation at zero, we swept the within-benchmark cross-half noise correlation from -0.1 to 0.5 with 1,000 replicates at each value. The mean margin estimate stays between 0.997 and 1.001 across that whole range, and no margin replicate reaches the observed 1.244. The correlation changes dispersion instead, which falls from 0.057 at zero to 0.046 at 0.5 on margins. Noise correlated this way adds the same quantity to the numerator and the denominator. It can't create inflation and instead pulls a real inflation toward one, which is the -0.014 bias that Table 14 records. On accuracy the null dispersion is wide enough that 0.19 of replicates reach the observed 1.078 with no correlation at all, and that share falls to 0.07 at a correlation of 0.5 . This sweep says nothing about noise shared across benchmarks within a half and repeated in the opposite half, which adds the same quantity to every entry of the cross-half covariance. The numerator absorbs all of it while the denominator absorbs only the diagonal tenth.

We measured that second violation in the same way as the first. Each half received a scalar effect that every benchmark loads on, repeated exactly in the opposite half. We swept its share of item-noise variance with 2,000 replicates at each value and no seed covariance. The mean margin estimate climbs monotonically from 0.999 at a share of zero to 1.606 at a share of 0.4 . A share near 0.124 reproduces the observed 1.244. At a share of 0.05 only 0.007 of replicates reach the observed value, so small amounts of this contamination don't explain the margin result. Accuracy is far more fragile, because the same construction reaches the observed 1.078 at a share near 0.009 , which is about 1 percent of item-noise variance. We haven't measured this component in the released scores. These figures give the contamination a reader would have to assume rather than a bound we established. The passage-aware split addresses shared structure inside one benchmark and leaves this cross-benchmark channel open. We tried to detect the contamination in the data and failed. Regressing the off-diagonal entries of the estimated seed covariance on the item-noise outer product gives a slope of 1.675 on margins, with a recipe bootstrap interval of 0.534 to 3.068, and 0.033 on accuracy with an interval of -0.006 to 0.073 . Simulations with the observed covariance and no shared effect give a mean slope of 1.682 on margins, and adding the shared effect at the share that reproduces the observed estimate moves that mean only to 1.808, against a replicate spread of 1.290 to 2.371. The statistic can't separate the two explanations, because a seed covariance that follows the item-noise scales looks the same as contamination that follows them. We therefore report that diagnostic as having no power rather than as a clean check.

One part of the assumption does admit a direct test, and we ran it. Write the per-item score of run r in configuration c as x_{cri} , and let μ_{ci} be its mean over that configuration's three runs, so that $d_{cri} = x_{cri} - \mu_{ci}$ carries the seed effect together with the item noise. Permuting d across the items of one group, separately for every run, leaves each group mean and therefore each trait score exactly where it was, and it leaves any component that all items of a group share exactly where it was as well, because permuting a constant returns the constant. What the permutation destroys is the tie between particular items and particular runs, which is the channel a shared passage or a shared source document travels through. Over 2,000 permutations of the 37,682 items in their 66 groups, the margin estimate has mean 1.2428 with a standard deviation of 0.0009, and 0.087 of the permutations reach or exceed the observed 1.244, with a Monte Carlo standard error of 0.006 on that share. On accuracy the permutation mean is 1.0763 with a standard deviation of 0.0053, and 0.352 of them reach or exceed the observed value. The item-level channel therefore accounts for about 0.001 of the margin estimate and 0.002 of the accuracy estimate, against excesses over one of 0.244 and 0.078. Re-drawing the item half split 200 times moves the estimate further on the real data than under permutation, since the real split standard deviation of 0.0018 on margins beats the 0.0009 the permuted data give, and 0.0083 on accuracy beats 0.0053. Item-level structure does survive, and it widens the spread without moving the centre. The shipped split returns 1.244 on margins against a re-split median of 1.246 and a range of 1.242 to 1.254, and 1.078 on accuracy against a median of 1.083 and a range of 1.059 to 1.103, so the reported numbers aren't a favourable draw. An effect that every item of a benchmark carries survives this permutation untouched. The whole-benchmark form of the violation stays untested and only the sub-benchmark form is bounded here.

The coverage table uses a generic generator, so we also built two scenarios fitted to these data, each with 2,000 replicates and a Monte Carlo standard error near 0.005. Both draw a per-band seed covariance from the nearest positive semidefinite matrix to the observed estimate in each size band, and both simulate 125 configurations with shared item halves. The second scenario also scales each recipe's seed deviations by the square root of that recipe's observed denominator sum divided by the mean recipe denominator. The first scenario doesn't reproduce the observed concentration, because the observed margin influence share of 0.520 exceeds 0.990 of its replicates, while the second puts that share at its 60th percentile. In the first, margin coverage is 0.954 for the wild interval, 0.946 for the cluster-robust t , 0.954 for the centred inversion, and 0.970 for the restricted inversion, with 0.947, 0.940, 0.951, and 0.966 on accuracy. In the second, the wild interval holds 0.946 and 0.945 while the cluster-robust t falls to 0.901 and 0.927. The centred inversion turns conservative at 0.977 and 0.958. We later reran both scenarios with two further constructions and a fresh seed, which reproduces every figure above inside two Monte Carlo standard errors and puts the wild interval at 0.946 and 0.942 on margins. The residual-scaled version keeps its advantage there, covering 0.960 in the first scenario and 0.964 in the second against those two. The alternative that the generic generator preferred also wins under a description fitted to these data. Correcting the cluster-robust t by $G/(G-1)$ doesn't rescue it, because it reaches only 0.939 and 0.909 on margins, which leaves the gap against the wild interval where it was. These scenarios take their covariance, their item noise, and their recipe weights from the same data that the intervals describe, so they show how each procedure behaves under a fitted description of this population and they aren't independent confirmation.

On the observed data the centred inversion gives 1.095 to 1.323 for margins and 0.995 to 1.162 for accuracy, beside wild intervals of 1.143 to 1.338 and 0.993 to 1.157. At the full battery the upper endpoints differ by at most 0.015 and the two constructions agree on which intervals include one, but the inversion puts the margin lower endpoint 0.048 lower. The wild lower endpoint shouldn't be read as a firm floor. In the size bands, the inversion gives 0.765 to 1.737 for 530M margins, which includes one although the wild interval excludes it, and 1.024 to 1.398 for 530M accuracy, which still excludes one. For 1B margins, where the wild lower endpoint is undefined, it gives a bounded set of 0.734 to 2.244. The accuracy intervals after BoolQ and WinoGrande removal stay above one under inversion, at 1.397 to 1.716 and 1.026 to 1.184.

The analysis also reports 1,000 trait-specific run-label permutations, each using 4,999 bootstrap draws, whose intervals contain the reference value one in 0.925 of margin permutations and 0.954 of accuracy permutations, falling below one in 0.049 and above one in 0.026 against nominal tail probabilities of 0.025. We repeated that experiment for this revision with 2,000 permutations, which gives coverage of 0.944 on margins and 0.949 on accuracy with a Monte Carlo standard error near 0.005, and tail shares of 0.039 below and 0.018 above. That looked like a two standard error gap

Table 14. Coverage of nominal 95 percent intervals across 2,000 simulated populations per cell. Bias is the mean estimate minus the population Λ . Noise gives the item-noise standard deviation relative to unit run effects. Inversion is the centred cluster test-inversion interval, computed later on the same simulated moments. Every cell was afterwards repeated at 10,000 replicates, and the main text quotes those ranges rather than the columns here, so the two differ by a few thousandths.

Population	Λ	Bias	Wild	Cluster t	Config.	Recipe	Inversion
Null, noise 0.73	1.000	0.000	0.953	0.946	0.948	0.937	0.953
Null, noise 1.45	1.000	-0.011	0.943	0.941	0.943	0.857	0.946
$\bar{r}_E = 0.061$, noise 0.73	1.245	0.001	0.947	0.943	0.938	0.929	0.948
$\bar{r}_E = 0.018$, noise 1.45	1.078	-0.007	0.943	0.940	0.942	0.910	0.945
One trait with 0.68 of trace	1.152	-0.001	0.938	0.936	0.937	0.925	0.941
One trait with 0.84 of trace	1.031	0.000	0.954	0.950	0.955	0.940	0.957
Recipe-shared run effects 0.5	1.245	-0.003	0.944	0.940	0.872	0.931	0.947
Student t , 4 df	1.245	0.000	0.946	0.930	0.943	0.927	0.955
One recipe at 6 times variance	1.245	-0.001	0.930	0.911	0.931	0.912	0.955
Cross-half correlation 0.1, noise 0.73	1.245	-0.014	0.928	0.926	0.930	0.918	0.932
Cross-half correlation 0.1, noise 1.45	1.078	-0.016	0.946	0.941	0.943	0.933	0.948
$\bar{r}_E = 0.2$	1.673	-0.002	0.947	0.943	0.943	0.928	0.948

from the record. We ran the experiment ten further times at the record’s own 1,000 permutations, each on a separate stream. Margin coverage across those ten runs falls between 0.915 and 0.942 with a mean of 0.931, and accuracy falls between 0.925 and 0.959 with a mean of 0.944. All four recorded coverages, including the cluster-robust ones of 0.881 and 0.941, land inside the matching range. The record reads as one draw at a replicate count too coarse to separate these values. The ten runs together put margin coverage 0.019 below nominal, which agrees with the 0.933 to 0.953 that the simulation gives. Across those ten runs the margin interval falls below the null in 0.032 to 0.052 of permutations, against a nominal 0.025 on that side. The lower-tail imbalance matters for interpreting an upper bound below the proposed threshold of 1.349, where the reported upper endpoint leaves only 0.011 of separation, approximately one quarter of a margin standard error.

These are conditional randomisation calibration results, contingent on the exchangeability assumed by the permutations. They aren’t coverage measurements for the unknown nonzero covariance of the original population. The cluster- t comparator has recorded coverage of 0.881 and 0.941 with above-one rates of 0.044 and 0.023, and our 2,000-permutation repeat gives 0.901 and 0.938 with above-one rates of 0.034 and 0.023. In the record and in the repeat alike, no margin estimate reaches the observation, while the accuracy upper-tail fraction is 0.021 in the record and 0.020 in the repeat, close to 0.028 in the earlier 500 draws. With a plus-one correction, zero exceedances among 1,000 draws give a Monte Carlo value of 1/1001, rather than zero.

The reported delete-one-recipe jackknife standard errors are 0.049 and 0.039. Applying a two-sided 95 percent $t(24)$ critical value to those rounded inputs gives approximate intervals of 1.143 to 1.345 for margins and 0.998 to 1.158 for accuracy. The corrected accuracy interval includes one, consistent with the primary interval.

Across the 1,000 permutations, estimator standard deviations are 0.0488 for margins and 0.0382 for accuracy. The earlier 500-draw values are 0.048 and 0.039, compared with N1 values of 0.046 and 0.048. The accuracy discrepancy therefore concerns simulated versus conditional permutation populations, rather than establishing an error in the observed bootstrap scale.

Competence proxy audit and calibration We audited the shipped competence adjustment and two repairs on the 375 reduced runs, then applied each adjustment to simulated populations with known seed covariance. The audit reproduces the shipped after-adjustment estimates of 1.244 and 1.076. The variance floor binds for WinoGrande on margins and for PIQA and WinoGrande on accuracy, and WinoGrande then supplies more than 99.99 percent of the squared standardised inputs to the margin proxy. The shipped margin adjustment in effect regresses every other benchmark on WinoGrande. Dropping each benchmark from the proxy in turn leaves margin inflation between 1.244 and 1.245, except for WinoGrande, whose removal gives 0.890 with interval 0.805 to 0.968.

The repair we fixed before seeing its output standardises by seed variances shrunk halfway toward their median positive value, estimated on four recipe folds and applied to the fifth. It gives margin inflation of 0.731 with interval 0.673 to 0.784, and accuracy inflation of 1.004 with interval 0.925 to 1.075. Cross-fitting without shrinkage leaves the floor binding and gives margin inflation of 1.245.

For this revision we fitted four proxy weightings out of fold, learning each variance, loading, and slope on four recipe folds and applying them to the fifth. A split-half check explains what the shipped proxy measures, since its half-A and half-B scores correlate at -0.086 on margins and -0.072 on accuracy. The shipped proxy carries no measurable signal about the run it scores. The shrunk proxy reaches 0.913 on margins and 0.301 on accuracy. A one-factor proxy with regression-score weights reaches 0.965 and 0.738. That factor puts 0.353 of its weight on one benchmark for margins and 0.467 for accuracy. Its fold weight vectors correlate at least 0.847 with each other.

Cross-fitted margin inflation is 1.245 under the shipped weighting, 0.731 after shrinkage, 0.973 for the factor proxy, and 0.889 when the clipped benchmarks are dropped, while the same four give 1.073, 1.004, 1.017, and 0.994 on accuracy. A recipe bootstrap of 999 draws, refitting the whole cross-fitted pipeline in each draw, gives 0.935 to 1.018 for the factor proxy on margins and 0.970 to 1.104 on accuracy, so the adjusted margin estimate stops separating from independence. Under simulated independent seed deviations all four variants return a mean between 0.998 and 1.003, so none of them manufactures covariance where none exists. None of this identifies mediation, because every one of these proxies averages the same seed deviations that it adjusts and so removes shared covariance by construction. Simulations settle what those cross-fitted numbers mean, since they supply the value each proxy returns when no competence variable exists. With the observed seed covariance and no competence variable at all, the four weightings return margin means of 1.097, 0.745, 0.980, and 1.029, and accuracy means of 1.109, 1.048, 1.066, and 1.072. The observed cross-fitted values sit inside the central 90 percent of those simulations for the shipped, shrunk, and factor weightings on both scales. None of them carries evidence of a competence mechanism beyond covariance removal. The clipped-drop weighting is the one departure, since its observed 0.889 on margins falls below a simulated fifth percentile of 0.908 and its observed 0.994 on accuracy falls below a fifth percentile of 1.003. Those simulations refit the cross-fitted pipeline in every replicate, so their percentiles describe a different estimator from the in-sample matched means that Table 15 reports for the same adjustment. Under a single common correlation of 0.178 the factor weighting returns 0.852 on margins, which is further below one than the matched simulations give. These proxies can overshoot when the covariance is equicorrelated.

Table 15 shows why none of these values measures competence, and under independent seed deviations every adjustment returns a mean between 0.999 and 1.002. The adjustments neither create nor remove covariance where none exists. With a single common correlation of 0.178, which reproduces the observed margin inflation, the repaired margin adjustments return 0.875 to 0.948, below one. With the observed seed covariance they return 0.745 to 0.889, each within 0.015 of the observed 0.731 to 0.890. On accuracy the repaired adjustments lower the matched simulated mean from 1.116 to between 1.049 and 1.079. The observed values of 1.004 to 1.019 lie inside the central 90 percent of those simulations. The repaired proxies average the same seed deviations that they adjust, so they remove shared covariance by construction and overshoot on margins. The matched simulations reproduce their observed values from the covariance estimate alone. A competence adjustment therefore needs a measurement taken outside the benchmark scores, such as held-out loss on the same checkpoints.

We then checked whether those checkpoints are reachable. An index built from the model repositories matched 120 of the 375 runs to the training branch whose released step set overlaps the run’s evaluation steps, with a median overlap of 1.0 and a smallest overlap of 0.042. Each of those 120 runs has a released checkpoint at its selected step. We built the remaining repository names from the evaluation tables, which call a recipe `dclm-baseline-top-10p` or `falcon-with-cc` or `dolma1.6++`, while the model repositories spell those same three as `dclm-baseline-qc-10p`, `falcon-and-cc` and `dolma1_6plus`. Seventeen of the 25 recipes are renamed that way, and because the Hub answers a request for a repository that doesn’t exist with HTTP 401 rather than 404, we read the refusals as gating. Resolving all 125 repositories under the released naming instead, every one of the 375 runs resolves without credentials to its own commit, each carrying a config, a tokenizer and safetensors weights that run from 0.762 GB at 150M to 5.119 GB at 1B. Among the 120 matched runs, the branch’s final step disagrees with the run’s final evaluation step in 21 cases, which is a further obstacle to scoring loss at the step the scores describe. The external measurement is therefore open to

anyone with the compute. The obstacle we reported earlier was our own naming rather than access control.

Table 15. Inflation after each competence adjustment, observed and in simulated populations with 300 replicates each. Simulated columns give replicate means under independent seed deviations, a single common correlation whose population inflation equals each scale’s observed value, and the nearest positive semidefinite matrix to the observed seed covariance.

Adjustment	Margin				Accuracy			
	Observed	Indep.	Common	Matched	Observed	Indep.	Common	Matched
None	1.244	1.000	1.242	1.243	1.078	0.998	1.093	1.116
Shipped, floored	1.244	1.000	1.170	1.093	1.076	0.999	1.092	1.109
Floored traits dropped	0.890	1.001	0.948	0.889	1.008	0.999	1.083	1.079
Shrunk, in sample	0.732	1.002	0.876	0.747	1.019	0.999	1.061	1.050
Shrunk, cross-fitted	0.731	1.002	0.875	0.745	1.004	0.999	1.060	1.049

Transport to PolyPythias The planned transport analysis scores nine runs per configuration on a nested 4,755-item battery. For three sizes, the source pipeline gives margin inflation of 1.40574 compared with 1.24395 for DataDecide, an excess ratio of 1.66323 computed from unrounded values. A scoring-kernel rewrite reproduces both inflation estimates to five decimals and reports the ratio as 1.66321, a difference that arises beyond the fifth decimal of the inputs. The configuration table requires the second-arm output rather than inspection of the DataDecide reduction alone.

On excess inflation, the DataDecide prediction falls approximately 40 percent below PolyPythias, while the raw inflation difference is approximately twelve percent. The planning document names aggregate standard deviation without resolving this scale choice, and the excess-based failure assessment is therefore post hoc. The raw inflation comparison falls inside the proposed twenty-percent band.

A later exploratory extension includes the 14M and 31M configurations, each with nine runs at step 143000 on the same 4,755 items. Within that rescore, margin inflation changes from 1.41372 at three sizes to 1.26199 at five sizes, while the excess ratio changes from 1.69592 to 1.07393. Accuracy changes from 2.28589 to 1.71085, compared with DataDecide’s 1.07837. Only the three-size accuracy point estimate exceeds the stated detection threshold of approximately 2.05. The configuration-bootstrap interval for five-size accuracy runs from 0.869 to 2.428, and at three sizes its lower endpoint is undefined, with an upper endpoint of 2.971.

The margin configuration-bootstrap intervals are 0.939 to 1.781 at three sizes and 0.927 to 1.537 at five sizes. Both contain one and DataDecide’s 1.24395, while the standard error falls from 0.190 to 0.150. These intervals limit conclusions about transport under either point-estimate assessment. The configuration bootstrap treats three or five configurations as enough to resample. We also inverted the centred cluster test of Appendix C with each configuration as its own cluster and $t(2)$ or $t(4)$ critical values. That construction gives margin sets of 0 to 2.106 at three sizes and 0.519 to 1.664 at five sizes. It can’t exclude any nonnegative accuracy value at either. These data don’t bound PolyPythias accuracy inflation at all. The rescore used one T4 with a 6,000-token cap instead of 30,000 tokens, and its three-size margin value of 1.41372 differs from the pipeline’s 1.40574 by 0.6 percent. The excess-ratio difference is approximately two percent, with hardware and batching changed together. The three-versus-five comparison holds those conditions fixed within one job. The source supplement describes 45 per-run score files from that job. For this revision we repeated the rescore on two T4 GPUs with the same scorer, token cap, items, and checkpoints. The repeat reproduces every estimate, configuration-bootstrap interval, and standard error that the earlier supplement reported, to the stated precision, and it corrects the five-size excess ratio, which the earlier supplement gave as 1.07395. We computed the inversion sets from the repeat’s own score files, because the earlier supplement didn’t report any inversion sets.

PolyPythias carries nine seeds for each configuration, which is enough to measure the ratio directly and to ask whether a three-run fit predicts it. The measurable quantity is the standard deviation of the battery mean over a set of runs, divided by the square root of the weighted sum of per-benchmark variances over the same runs, and it includes item noise. For each of the 84 ways of choosing three

runs, we formed three predictions of that quantity on the six runs left out. Independence predicts one, while the plug-in predicts the inflation estimate itself, and the decomposition adds each benchmark’s estimated item noise to both sides of the ratio. On margins across three sizes the mean absolute log error is 0.339 for independence, 0.536 for the plug-in, and 0.512 for the decomposition, so neither fitted prediction beats the constant. On accuracy the same three errors are 0.370, 0.447, and 0.384, while the decomposition beats independence in 0.50 of splits and the five-size set repeats this ordering. The decomposition beats the plug-in in 0.81 and 0.75 of margin splits, which is what the item-noise term is meant to do.

The six-run target is itself noisy, because the measured margin ratio runs from 0.70 to 1.85 between its fifth and ninety-fifth percentiles. This compares predictions of one noisy realisation rather than of the underlying ratio. Within that limit, the decomposition doesn’t show out-of-sample value against a single configuration’s ratio.

Pooling the target across configurations changes the answer, which shows how much of the first comparison was target noise. When the six held-out runs give one ratio built from mean within-configuration variances, its standard deviation across splits falls to 0.140 on three-size margins. The decomposition then has mean absolute log error 0.275 against 0.319 for independence on three-size margins, and 0.367 against 0.523 on accuracy. It wins in 0.60 and 0.73 of splits. The five-size set gives 0.209 against 0.218 on margins and 0.251 against 0.364 on accuracy. The decomposition also beats the plug-in on every set, by between 0.010 and 0.105 of log error, so carrying item noise into the prediction helps on all four sets, where the covariance term helps on three of them and hurts on five-size margins. The 84 splits share runs and the four comparisons share configurations, so these are descriptive differences without an interval.

Table 16. Five-fold prediction error for log aggregate seed standard deviation. Every model receives held-out trait standard deviations, so the comparison measures correlation prediction with variance estimation fixed.

Model	Parameters	Margin error	Accuracy error
P0 independence	K	0.0358	0.0143
P1 rank-one competence	$K + 1$	0.0057	0.0140
P1G rank-one plus gain	$2K + 2$	0.0063	0.0260
P2 performance correlation	K	0.1073	0.1682
P3 full covariance	$K(K + 1)/2$	0.0068	0.0105

Table 16 gives fold-mean point errors, and Appendix .6 gives paired intervals for their differences. P1 and P1G have lower point errors than P3 on margins and higher errors on accuracy, while P0 and P2 have higher errors on both scales. P2 uses performance correlation from the held-out fold itself because its calculation needs no seed replication. Negative held-out variance estimates are clipped to zero before model selection, which also applies to independence. On the complete population this clips one margin and two accuracy traits, increasing the aggregate covariance sum by 0.3 and 4.8 percent, while the scored target retains the original estimates. Shared clipping doesn’t guarantee that it leaves model rankings unchanged. We repeated the comparison under two alternative rules, with the same 2,000-draw paired bootstrap as Appendix .6. On margins, the variance is nonpositive for OpenBookQA and WinoGrande in one fold and for WinoGrande in two more. Dropping an affected trait from that fold’s prediction, correlation, and target gives margin errors of 0.0063 for both P1 and P3, and filling its held-out variance with the training-fold estimate gives 0.0039 and 0.0045, while independence stays near 0.035. Under both rules the P1G and P3 intervals for a difference from independence still exclude zero, but under training fill the P1 interval runs from -0.059 to 7×10^{-5} and so includes zero. On accuracy, one to three traits are affected in every fold, and dropping them leaves P3 the best model, with its error falling from 0.0105 to 0.0090, while filling gives P1 and P3 errors of 0.0122 and 0.0118. None of the accuracy intervals for these contrasts excludes zero under either rule.

Gain calibration and recipe split sensitivity The default gain simulator applies a shared margin scale across traits. The matched simulator instead copies each benchmark’s mean margin, item-noise variance, and seed variance from the data, and it keeps seed deviations independent across benchmarks. At $\sigma_g = 0.05$, the two reported excess shares differ by a factor of about 7.5. Subtracting

estimated leakage variance 1.91×10^{-3} from total gain variance 3.67×10^{-3} gives a gain standard deviation of approximately 0.042. This subtraction depends on the variance decomposition assumed for the covariate.

Table 17. Gain-simulator sensitivity relative to the observed margin excess of 0.244. These shares compare specified simulations and don’t bound a causal contribution in the observed runs.

σ_g	Default $\hat{\Lambda}_M$	Excess share	Matched $\hat{\Lambda}_M$	Excess share
0.01	1.007	0.029	1.001	0.003
0.02	1.026	0.108	1.003	0.014
0.03	1.058	0.236	1.008	0.031
0.042, estimated	1.108	0.443	1.014	0.059
0.05, default	1.149	0.609	1.020	0.081
0.10	1.465	1.905	1.062	0.256

The default 0.05 cell differs by approximately 0.0009 from the corresponding Table 9 result before rounding, less than half the reported Monte Carlo standard error for 500 replicates. The G5 assessment uses that default specification, which fails its threshold. Accuracy estimates stay within 0.004 of one in these simulations, but positive multiplication preserves signs by construction. Their proximity to one therefore doesn’t establish absence of a gain mechanism for observed accuracy variation.

To carry uncertainty in the gain scale into these shares, we resampled the 25 recipes 999 times and re-estimated the gain standard deviation in each draw. The median is 0.041, the 95 percent percentile interval runs from 0.011 to 0.063. The leakage-corrected variance is nonpositive in 0.9 percent of draws. With 300 replicates at each endpoint, the default simulator gives margin inflation of 1.006 and 1.221, which are shares of 0.02 and 0.91 of the observed excess, while the matched simulator gives 0.999 and 1.028, with shares of -0.01 and 0.12. At the point estimate, 300 new replicates give 1.105 and 1.012, within two Monte Carlo standard errors of the 500-replicate values in Table 17.

We also compared each simulator’s populations with the observed runs at the point estimate. The default simulator works on a standardised scale, with seed variances between 1.02 and 1.05 and mean margins near 3.85, whereas the observed seed variances lie below 0.002 and the observed mean margins lie between -0.18 and 0.07. Its shares depend on a ratio of margin level to seed noise that it assumes rather than measures. The matched simulator reproduces the observed mean margins and item-noise variances of every benchmark to within 1.2 percent, and the seed variances of the four benchmarks with the largest estimates to within 6.1 percent. Its margin configuration means vary far less than the observed ones, with standard deviations between 0.02 and 0.45 times the observed values, because it adds no systematic differences between configurations. Its accuracy seed variances exceed the observed estimates by factors of 2.4 to 116 on the eight benchmarks where those estimates are positive. When we apply the gain estimator to the simulated populations, it recovers a mean of 0.037 under the default simulator and 0.036 under the matched simulator, 11 and 15 percent below the input of 0.042. The estimator itself understates the gain scale in both specifications.

For a recipe subset S , the estimator is $\hat{\Lambda}_S = \sqrt{\sum_{c \in S} T_c / \sum_{c \in S} U_c}$, because each configuration’s components require no other configurations, and the analysis enumerates all $\binom{25}{17} = 1,081,575$ possible seventeen-recipe estimation sets. It reports completing this sweep in thirteen seconds on one CPU core. The reconstructed split uses the plan’s master seed under the released implementation, but the plan didn’t specify a complete reconstruction rule and no contemporaneous split artifact was recovered. Table 18 summarises that exhaustive sweep over every admissible estimation subset.

No subset reaches either proposed point threshold in this enumeration. The sweep measures sensitivity to recipe selection after the analysis choices were made. It provides no check on choices that used the full dataset.

That enumeration moves the point estimate, and it doesn’t say whether the reported conclusions themselves survive a deletion. We recomputed the shipped interval after removing each recipe, each size band, and each benchmark in turn. For the full-battery margin we also removed all 300 pairs of recipes. The margin interval keeps excluding one under every one of the 25 single-recipe deletions,

Table 18. Exhaustive sensitivity to the choice of seventeen estimation recipes. Every admissible subset gives a finite point estimate on both score scales.

Quantity	Margin	Accuracy
All 25 recipes	1.244	1.078
Reconstructed planned partition	1.199	1.099
Minimum	1.037	0.963
5th percentile	1.176	1.035
Median	1.246	1.078
95th percentile	1.285	1.124
Maximum	1.317	1.191
Effective-count range	[5.77, 9.31]	[7.05, 10.77]
Planned threshold	1.349	1.40
Subsets reaching threshold	0	0
Subsets below one	0	1,393, or 0.13%

under all 300 pairs, and under every single-benchmark deletion, where the estimate over the pairs stays between 1.177 and 1.263 and the nearest endpoint sits 0.037 above one. One deletion breaks it, because dropping the 530M band leaves 1.178 with an interval of 0.993 to 1.338. The lower endpoint falls 0.007 below one on the remaining 100 configurations. We calibrated that exception the same way we calibrate the accuracy side. Simulating 10,000 margin-like populations over two runs with separate seeds, and keeping the 9,697 whose full interval excludes one, at least one of the five band deletions loses the exclusion in 0.164 of them, and at least one of the 25 recipe deletions loses it in 0.041. The two runs agree on the band figure inside a quarter of a standard error of their difference and on the recipe figure inside one and a quarter. The later run carried 8,000 of those replicates. One band failure out of five is therefore close to what a deletion sweep of this design produces by chance, which is a weaker statement than the deletion looks and a fairer one.

The same simulator answers the question a reader planning their own replicate study would ask, which is how much data an effect of this size needs. At our 125 configurations and three runs the margin effect clears one in 0.970 of replicates while the accuracy effect clears it in 0.111. The two conclusions differ in power before they differ in substance. Holding three runs and raising the recipe count to 50, 100 and 200 takes the accuracy figure to 0.165, 0.306 and 0.620. Six runs per configuration take the same four designs to 0.196, 0.398, 0.704 and 0.947. Four fifths therefore needs between 500 and 1,000 configurations at six runs, which is eight to sixteen times the run budget of this release, and that number is the honest answer to why our accuracy interval includes one.

One deletion remains that none of this covers, which is the deletion of a run. Each configuration holds three of them, so a single anomalous run has less data standing between it and the estimate than any column of the design. Re-estimating on each surviving pair leaves the margin estimate at 1.256, 1.242 and 1.235, all with intervals that exclude one, and the accuracy estimate at 1.084, 1.114 and 1.030, none of which excludes it. Because two runs leave one contrast direction instead of two, the margin interval widens from 0.195 to between 0.201 and 0.356, which is the cost of that lost direction rather than a change in the answer.

The accuracy conclusion is the fragile one, and it is fragile in the opposite direction, because its full-battery interval includes one yet stops including it under five of the 25 recipe deletions, under the removal of the 750M band, and under the removal of WinoGrande as well as BoolQ. All eight of those exceptions put the whole interval above one rather than below it, seven of them with lower endpoints between 1.002 and 1.025. The eighth is the BoolQ removal that we already report at 1.412.

Twenty-five correlated deletions of the same data give plenty of chances to flip something. We calibrated the count against a null rather than reading it directly. Simulating 10,000 populations whose true ratio is exactly one under accuracy-like noise, again over two runs with separate seeds, and keeping the 9,477 whose full interval includes one, at least one recipe deletion flips in 0.105 of them and five or more flip in 0.018. A second null calibrated to the observed 1.078 puts those at 0.127 and 0.032 over its 8,964 kept replicates. All six of these calibration figures reproduce across

the two runs inside 1.8 standard errors of their difference. The larger repeat sharpened them without changing what they say. The observed five recipe flips therefore sit in the upper few percent of what a deletion sweep produces by chance, which is why we read them as a direction rather than as noise. The deletions that move the accuracy conclusion move it towards more inflation. A reader should treat the inclusion of one on the full battery as a statement about that battery rather than about accuracy in general.

Because one size band carries that much weight, we also asked whether clustering on recipes alone leaves the interval too narrow. Clustering the same linearised residual on the five size bands instead gives a standard error of 0.175 on the squared margin ratio against 0.112 from the recipes. Its interval runs from 1.030 to 1.426 with four degrees of freedom and a critical value of 2.776, so it still excludes one. The matching accuracy interval runs from 0.961 to 1.184 and still includes one, which means both full-battery conclusions survive a clustering on the dimension the deletions flagged.

The two-way variance of Cameron, Gelbach and Miller doesn't help here, because the component for the intersection of recipe and size band, which puts each configuration in its own cluster, gives 0.233 and exceeds the sum of the other two. The two-way variance turns negative on the full-battery margin and no interval exists. On full-battery accuracy the same subtraction returns a standard error under half the recipe one, which we don't believe and don't report as a result. The estimator is better behaved on the battery without BoolQ, where it widens the recipe standard error by factors of 1.29 and 1.40 and leaves both scales above one, at 1.652 to 1.910 and 1.276 to 1.796.

A bootstrap version of the same correction has no negative-variance failure mode, because it multiplies each residual by the product of one Rademacher draw per dimension and reads the interval off the resulting spread, so we ran it at 9,999 draws. It returns 1.051 to 1.411 on the full-battery margin, which is the cell the analytic version couldn't deliver, and it still excludes one. On full-battery accuracy it returns 0.973 to 1.175, which is wider than the recipe interval rather than narrower. The analytic accuracy figure we declined to report was the artefact we took it for.

The bootstrap doesn't repair the failure mode itself. We first measured that over 1,000 replicates at 999 draws, which left both comparisons inside their own Monte Carlo noise. We repeated the two cells at 8,000 replicates and 2,999 draws and scored the two intervals on the same replicate each time. At a band-shared quarter the two-way bootstrap covers 0.823 against 0.833 for the recipe-clustered interval, and at the most extreme joint cell the data allows, it covers 0.932 against 0.940. Because the two intervals see identical data, the paired difference carries the comparison. It comes to -0.010 with a standard error of 0.0024 at the quarter and -0.008 with a standard error of 0.0019 at the corner. Both sit more than four standard errors below zero. The two-way interval is measurably worse here rather than merely no better, and we therefore keep the recipe clustering. The repeat also reproduces the four observed cells at 9,999 draws to every digit we report. We note that five size clusters are too few for the analytic two-way correction at this design.

The released seed labels make the failure mode of that choice concrete, because the release uses seeds 2, 14 and 15 below 1B and seeds 2, 4 and 5 at 1B, so one seed label is shared across every recipe inside a size band. We checked that those labels mean what we take them to mean, rather than assuming it. The model repositories name their branches by step and by one of default, small-aux-2 and small-aux-3, and nothing in the release states which released seed each branch produced. Rescoring the first 150 ARC-Easy documents at 150M from the request file the release itself ships, and forming the same per-byte margins, matches the default branch to released seed 2, small-aux-2 to seed 14 and small-aux-3 to seed 15. It does so in all eight recipes our first index resolved, which are c4, falcon, fineweb-edu, fineweb-pro, dclm-baseline and its three blends with dolma 1.7, giving 24 branch cells without a single exception. Twenty of those matches carry a correlation of 0.99999997 or better with a median absolute difference between $1.0e-05$ and $2.0e-05$, and the four weakest still reach 0.99993. The best wrong seed in any cell is 0.897, so no cell is ambiguous. A separate run repeats the check at 1B, where it covers five recipes. The 1B band is where the release changes both halves of the naming, shipping seeds 2, 4 and 5 and branches called large-aux-2 and large-aux-3. It is the band where a silent difference would have hurt most. Rescoring the same 150 documents there matches the default branch to seed 2, large-aux-2 to seed 4 and large-aux-3 to seed 5 in all fifteen cells it loaded, without an exception, and those fifteen cover five of the eight recipes that resolve at 1B rather than all of them. The weakest of those fifteen correlations is 0.99986 and the best wrong seed in any cell is 0.910, and no cell is close to ambiguous. The scores we analyse

therefore reproduce from public weights. The seed labels that this clustering argument leans on are the labels we believe they are wherever we could check them, although the other seventeen recipes went unchecked because we had misnamed their repositories.

A direct census of all 125 repositories looked at first like two reasons a load can fail. Eighty answered an unauthenticated request with HTTP 401, five carried the name `dolma1.6` with two plus signs, which the hub rejects as an identifier, and 40 resolved. That split was identical in each of the five size bands, at sixteen refusals, one rejected name and eight open repositories, and a repeat census six and a half hours later returned the same 80, 40 and 5. The regularity was the clue we missed, because access control has no reason to fall on the same sixteen recipes in every band. Those 85 names came from the evaluation tables. The model repositories rename seventeen of the 25 recipes, writing `qc` where the tables write `top`, and `-cc` where they write `with-cc`, and `dolma1_6plus` for `dolma1.6++`. An anonymous client can't tell a private repository from a missing one. The Hub answers both with 401, and our census recorded that code for names that never existed. Under the released names all 125 repositories resolve without credentials, every one of the 375 runs carries its own commit with a config, a tokenizer and weights, and the branch list reads `default`, `small-aux-2` and `small-aux-3` below 1B and `default`, `large-aux-2` and `large-aux-3` at 1B.

We simulated a run effect that follows the size band rather than the recipe, built exactly as the existing recipe-shared effect is built so that the true ratio doesn't move. We swept its share of latent variance with 2,000 replicates at each value, then repeated the three shares the limitation rests on at 10,000 more with a second seed. At a share of zero the recipe-clustered wild interval covers 0.943 and the size-clustered one covers 0.951 over the pooled 12,000, so the two agree when the assumption holds. At a quarter the recipe-clustered interval falls to 0.821, at a half it reaches 0.580, and at three quarters, which only the first pass ran, it reaches 0.348, while its median width shrinks from 0.242 to 0.189 because the recipe clusters never see the added variance. The size-clustered cluster-robust t holds 0.935, 0.898 and 0.758 over the same sweep. The two passes agree to within 0.013 at every cell, and the pooled figures are the ones we quote. A reader who thinks this dependence is present should therefore read the size-clustered interval on the observed data, which runs from 1.030 to 1.426 on margins and still excludes one, and from 0.961 to 1.184 on accuracy and still includes one. Both reported conclusions survive the clustering that is robust to this violation, but the reported interval's width doesn't.

Recommending that fallback obliges us to test it against the sharing channel these data actually bound, which is the recipe one rather than the band one. We swept a recipe-shared run effect over 0, 0.114, 0.25 and 0.50 at 4,000 replicates with seed 20260955. The size-clustered wild interval covers 0.953, 0.955, 0.944 and 0.898 across that sweep, and its cluster-robust counterpart covers 0.947, 0.944, 0.929 and 0.869. At the 0.114 bound the permutation test puts on recipe sharing, the fallback therefore costs nothing at all. It only starts to slip at a half, which sits more than four times that bound. The accuracy scale changes which version of that fallback a reader should use. Repeating the band-sharing sweep at the accuracy battery's correlation and noise leaves the recipe-clustered interval at 0.950, 0.947, 0.943, 0.916 and 0.801 across shares of 0, 0.044, 0.114, 0.25 and 0.50. The damage arrives later there than on margins, where a quarter already costs 0.821. The size-clustered wild interval is the one that breaks on accuracy, covering only 0.832 even when no sharing is present, while its cluster-robust counterpart holds 0.935. Five clusters carrying the louder item noise are too few for the bootstrap, and a reader taking the fallback on accuracy should take the cluster-robust version. Sweeping recipe sharing on the same scale confirms that the failure belongs to the cluster count rather than to the sharing, since the size-clustered wild interval sits between 0.830 and 0.838 at every share from zero to a half while its cluster-robust counterpart holds between 0.916 and 0.936. The recipe-clustered interval the paper reports stays between 0.936 and 0.948 throughout. Those two sweeps used seeds 20260983 and 20260984 at 4,000 replicates. The fallback costs width on the margin scale instead, since in the recipe-sharing sweep under seed 20260955 five clusters leave a median width near 0.357 against 0.246 for the recipe-clustered interval on the same replicates. The recipe-clustered interval in that margin sweep holds between 0.942 and 0.947 throughout, which is what a sweep of the dependence it already models should show. The reimplemented interval reproduces the shipped one exactly in every cell.

Two sweeps ask whether any of this survives a design that isn't the one the release happens to ship. Widening the battery from four benchmarks to twenty raises the true ratio from 1.088 to 1.469, because flat weights over a compound-symmetric truth put the squared ratio at one plus k minus one times the correlation. The recipe-clustered interval covers between 0.941 and 0.951 at every width

with no undefined replicate anywhere. Varying the runs per configuration from two to ten leaves that interval between 0.942 and 0.952, while the median width falls from 0.343 at two runs to 0.242 at the three the release ships and 0.114 at ten. The size-clustered fallback is the one that suffers at two runs, dropping to 0.915. A reader with only two runs per model should keep the recipe clustering. Both sweeps ran at 4,000 replicates with seeds 20260956 and 20260957. A companion sweep varies the number of size bands rather than the runs, holding the cell count near 125 by pairing 41 recipes with three bands, 25 with five, 16 with eight and 10 with twelve. The recipe-clustered interval covers between 0.938 and 0.948 across all four. The band count doesn't reach the clustering the paper reports. The fallback is the arrangement that cares, since the size-clustered wild interval covers only 0.900 with three bands and recovers to between 0.943 and 0.947 at five, eight and twelve, which says the fallback needs at least the five bands this release ships. A third sweep moves the item noise itself over 0.25, 0.50, 0.73, 1.00, 1.45, 2.00 and 3.00, which brackets both scales the paper reports and runs past them. The recipe-clustered interval covers between 0.944 and 0.954 from the quietest cell through 2.00. It collapses at 3.00 to 0.668 with 0.312 of replicates undefined. The estimator breaks somewhere near four times the margin battery's noise rather than anywhere near the accuracy battery's 1.45. Power is what the noise actually buys, because the exclusion rate at a true 1.2446 falls from 0.998 at the quietest cell to 0.630 at 1.45 and 0.326 at 2.00. These two ran at 4,000 replicates with seeds 20260977 and 20260976.

Those sweeps moved one noise level for the whole grid. A real battery scores less consistently at some model sizes than at others. A further run lets the item noise vary by size band. Spreading it two-fold across the five bands leaves the recipe-clustered interval at 0.945 against 0.945 for a flat grid, while the median width grows from 0.241 to 0.310 and the exclusion rate at a true 1.2446 falls from 0.968 to 0.825. A four-fold spread leaves coverage at 0.935 with the loud band last and 0.938 with it first, and the direction of the ramp carries nothing. An alternating pattern costs a little more at 0.919. What breaks the estimator is the loudest band rather than the unevenness, because a four-fold ramp on margins takes its noisiest band to 2.92 and holds coverage, while the same ramp at the accuracy battery's base noise takes it to 5.80 and coverage falls to 0.208 with 0.776 of replicates undefined. A single loud band tells the same story, holding 0.947 on margins and falling to 0.291 on accuracy. A sixteen-fold ramp on margins, whose noisiest band reaches 11.68, takes coverage to 0.111 with 0.885 of replicates undefined. The threshold rather than the spread is what a reader has to check. A reader should therefore compare their noisiest band against the collapse near 3.00 that the previous sweep found rather than ask how uneven their bands are (seed 20260991, 4,000 replicates).

Every sweep so far hands the estimator all 125 configurations, and a practitioner rarely has a complete grid. A fourth run removes cells at random and measures what the loss costs. Dropping a tenth, a quarter, a third and a half of the configurations leaves the recipe-clustered interval covering 0.950, 0.940, 0.940 and 0.945, and removing six whole recipes rather than scattered cells leaves 0.945 on nineteen clusters. Cluster balance therefore doesn't matter here, which is the opposite of what cluster-robust inference usually warns about. The loss shows up in width and power instead. The median width grows from 0.242 to 0.342 as half the grid disappears. The exclusion rate at a true 1.2446 falls from 0.968 to 0.742. A reader missing half their cells keeps a trustworthy interval and loses a quarter of their chances of resolving the question. The accuracy scale pays slightly more for the same loss, covering 0.949 complete, 0.942 at a quarter dropped and 0.931 at a half, where 2 percent of replicates also stop being defined, while a lumpy quarter again costs nothing at 0.947. Those two runs used seeds 20260974 and 20260981 at 4,000 replicates.

Those drops remove whole configurations, and the release also loses single seeds where a branch stops early or misses the shared step. A further run draws the run count per configuration rather than fixing it. Cutting a tenth, a quarter, a half and then all of the configurations to two seeds leaves the recipe-clustered interval at 0.946, 0.944, 0.948 and 0.948 with no undefined replicate anywhere, and cutting whole recipes rather than scattered cells leaves 0.946 and 0.939. The accuracy scale behaves the same way at 0.949 and 0.947. The shortfall is paid in width and power, because the median width grows from 0.243 to 0.343 and the exclusion rate at a true 1.2446 falls from 0.968 to 0.747 as every configuration drops to two seeds. Unevenness costs something even when it is free, since a mix of two, three and four seeds averaging exactly three per configuration gives a median width of 0.265 against 0.243 for a uniform three. A reader with a fixed budget should spread their seeds evenly. That run used seed 20260989 at 4,000 replicates.

The recommendation to measure this ratio locally is worth a price tag. A last design run shrinks both dimensions at once, to the ten to twenty five recipes and two to four runs a small lab can afford.

Coverage survives that squeeze, since the recipe-clustered interval covers 0.936 at ten recipes and two runs, 0.945 at fifteen and two, and between 0.942 and 0.949 everywhere else in the grid. Power is what a cheap design gives up, because the exclusion rate at a true 1.2446 runs 0.266 at ten recipes and two runs, 0.449 at fifteen and two, 0.542 at ten and three, and 0.736 at the shipped twenty five with two. Only twenty five recipes with four runs reaches 0.997, which sits above the 0.968 the shipped three-run design achieves. A reader planning their own measurement should therefore budget for recipes first and runs second, and should read a null result from a ten by two design as uninformative rather than as independence (seed 20260975, 4,000 replicates).

The accuracy scale makes that advice sharper, because its cheapest corner doesn't work at all. Ten recipes with two runs cover 0.602 with 0.369 of replicates undefined, and ten with three reach 0.859. The design the margin grid still tolerated at 0.936 collapses at this noise. Coverage returns once either dimension grows, reaching 0.933 for ten recipes with four runs, 0.942 for fifteen with three, 0.951 for twenty with three and 0.943 for the shipped 25 with two. The requirement is therefore joint rather than a recipe count on its own. A reader at this noise needs fifteen recipes with three runs or 25 with two. Power stays out of reach across the whole grid, since the exclusion rate at a true 1.078 never passes 0.142 even at 25 recipes with four runs (seed 20260993, 4,000 replicates).

A third sweep asks how many recipes the clustering itself needs, because 25 clusters is a small number for cluster-robust inference and the wild bootstrap is the standard answer to exactly that problem. Sweeping the recipe count over 5, 10, 15, 25, 40 and 60, the wild interval covers 0.749, 0.946, 0.942, 0.942, 0.946 and 0.946. The shipped design sits well above the failure point rather than on it. The failure at five clusters is worth naming, since there the wild interval does worse than the plain cluster-robust one, which holds 0.910, and the usual advice would have predicted the opposite. The extra clusters buy power rather than coverage, because the exclusion rate at a true ratio of 1.2446 climbs from 0.54 at ten recipes to 0.97 at the shipped 25 and 1.00 at sixty, and that sweep ran at the margin battery's item noise. The accuracy battery carries twice it, so we repeated the same six cluster counts there and the requirement moves. The wild interval covers 0.360 at five recipes with 0.606 of replicates undefined, then 0.869 at ten, 0.943 at fifteen, 0.960 at the shipped 25, and 0.949 and 0.950 at forty and sixty. Ten recipes are therefore enough on margins and short on accuracy. A reader working at that noise should treat fifteen as their floor rather than ten. The plain cluster-robust interval tracks the wild one within 0.003 at every count from fifteen upward, which makes the cluster count matter far more than the choice between the two constructions. The size-clustered fallback recovers as the grid grows rather than as its own cluster count grows, since it holds five clusters throughout while covering 0.850 at 25 recipes, 0.911 at forty and 0.942 at sixty. Its failure came from thin clusters rather than from few ones. Power stays the binding constraint on this scale, because the exclusion rate at the true 1.078 climbs only from 0.095 at 25 recipes to 0.207 at sixty (seed 20260987, 4,000 replicates).

Two further sweeps break the assumptions the generator has been handing the estimator all along. Drawing the latent run effects from a multivariate t leaves coverage at 0.944 with ten degrees of freedom and 0.940 with five, and only a t with three, which has no finite kurtosis, pulls it down to 0.895. Giving the 25 recipes variance scales spread over a four-fold and a sixteen-fold range leaves coverage at 0.938 and 0.946, so unequal recipes cost nothing measurable. The last sweep prices the assumption the whole estimator rests on, which is that the two item halves carry errors that don't correlate. A correlation of 0.02 or 0.05 between them leaves coverage at 0.946 and 0.943, a correlation of 0.10 costs about a point at 0.928, 0.20 costs two at 0.920, and 0.40 costs seven and a half at 0.864. Our passage-level check moved the estimates by less than 0.001, which places the real leak far below the first of those steps. The sweep says what a larger one would have cost. All three ran at 4,000 replicates with seeds 20260960, 20260961 and 20260962.

Two more sweeps finish the picture of what this design can stand. Giving each benchmark the item noise its size implies, from the item counts the release ships, which spread from 500 on Open-BookQA to 14,042 on MMLU, leaves coverage at 0.945 against 0.946 for equal noise, and doubling that spread leaves it at 0.948, so the equal-noise simplification cost nothing. Crossing three bootstrap weight families with the cluster count says the same about the weight choice, since at the shipped 25 recipes Rademacher, Mammen and Webb weights cover 0.948, 0.939 and 0.948, and they only separate at five clusters where all three have already failed. Mammen weights do best in that failed corner at 0.873 against 0.830 for Webb and 0.760 for Rademacher, which is a curiosity rather than a recommendation, because no reading of a 0.873 interval is safe. Those families are the bootstrap's own weights, and the weights a reader gives the benchmarks matter more. A further run replaces the

flat one tenth with three alternatives. Weighting the benchmarks by the item counts the release ships, which run from 500 to 14,042, putting them on a ramp whose last benchmark carries four times the first, and concentrating half the weight on two benchmarks all leave the recipe-clustered interval covering between 0.941 and 0.951 against 0.947 for flat weights. The weighting moves the quantity being estimated rather than the coverage of it, because the true ratio falls from 1.2446 under flat weights to 1.2128 on the ramp, 1.1530 under concentration and 1.0975 under item counts, and the exclusion rate falls with it from 0.962 to 0.509. A reader who aggregates by item count is therefore asking a question this design would resolve about half the time. The interval they read would still be honest (seed 20260986, 4,000 replicates).

The last question a reader will have is what the within-size evidence would have needed, and the answer is runs rather than cleverness. A single band of 25 configurations at the shipped three runs covers 0.930 while excluding one in only 0.313 of replicates at a true ratio of 1.2446, which is why the 1B interval includes one, and six runs lift that to 0.739 and ten runs to 0.957 with the median width falling from 0.543 to 0.253. Ten runs per configuration inside one band would therefore have settled the question the paper has to leave open. These ran at 4,000 replicates with seeds 20260966, 20260967 and 20260963.

Every sweep above hands the estimator a truth where all ten benchmarks share one correlation, and this battery isn't built that way. The last run measures the real covariance and sweeps against that instead. Ten single-benchmark restrictions give the diagonal and forty five pairs give the off-diagonal, all of it read through the shipped estimator rather than by hand. The matrix that comes back reproduces the reported ratios to four decimals, at 1.2439 on margins and 1.0784 on accuracy. Projecting it onto the positive semidefinite cone clips one small negative eigenvalue on margins and three on accuracy, moving the ratios to 1.2438 and 1.0778. Simulating against those measured matrices leaves the recipe-clustered interval covering 0.947 on margins and 0.945 on accuracy. The flat truth the other sweeps use was a harmless simplification rather than a flattering one. That run also prices the accuracy conclusion, because at a true ratio of 1.0778 the interval excludes one in 0.103 of replicates, which sits 1.6 Monte Carlo standard errors from the 0.111 the compound-symmetric design power run reports at the same fitted cell. A separate sweep of the true ratio at the accuracy battery's noise says the same thing in more detail, since the exclusion rate runs 0.018 at a ratio of exactly one, 0.063 at 1.05, 0.135 at 1.10, 0.281 at 1.15, 0.668 at 1.25 and 0.968 at 1.40. The accuracy battery could therefore only have found dependence reliably somewhere near 1.35. A reader should treat its interval as evidence against a large ratio rather than as evidence for independence. Both ran at 4,000 replicates with seeds 20260965 and 20260964.

The margin battery is the more sensitive of the two by a factor near two and a half, since at its own item noise the exclusion rate runs 0.014 at a ratio of one, 0.104 at 1.05, 0.332 at 1.10, 0.652 at 1.15 and 0.970 at 1.25. The point estimate itself is close to unbiased across that whole range, which matters because a ratio of sums under a square root has two reasons to run low. Over the same sweep the mean estimate sits between 0.0005 above and 0.0028 below the truth on margins, and between 0.0031 and 0.0067 below it on accuracy, against estimate standard deviations of 0.055 and 0.098. The bias is therefore under a tenth of a standard deviation everywhere. No replicate in either sweep produced an undefined estimate. Coverage holds across that range too, since sweeping the true ratio at the margin battery's item noise leaves the recipe-clustered interval at 0.944 when the truth is exactly one, then 0.934, 0.939, 0.949, 0.946 and 0.943 at 1.05, 1.10, 1.2446, 1.40 and 1.60. The cell at a true ratio of one is the one worth naming, because there the interval excludes one in 0.018 of replicates, which is the false positive rate a reader should assume when they apply this estimator to a battery with no dependence at all. That sweep used seed 20260985 at 4,000 replicates. The bias sweep ran at 4,000 replicates per cell under seed 20260973.

A last pair of runs tests the two things those sweeps still take on trust. Five explicit covariance shapes, a flat one, one giving a single benchmark 68 percent of the trace, one adding negative links from that benchmark to the other nine, one splitting the battery into two correlated blocks, and one combining dominance with negative links, leave coverage between 0.936 and 0.946 over 4,000 replicates. The flat shape looked like the weak one there, at 0.9357 against 0.9463 for the two blocks. We repeated it and the dominant shape at 12,000 replicates with a fresh seed, and the ordering reverses. Flat covers 0.9471 and dominant covers 0.9429 on the larger run, a gap of 1.4 Monte Carlo standard errors. The shapes agree and the first reading was noise (repeat seed 20260982). Two of those shapes carry a true ratio below one, at 0.9927 and 0.9900. The interval excludes one in 0.011 and 0.009 of replicates there. The estimator doesn't manufacture dependence when the truth runs

the other way. The permutation null earns the word exact under the same abuse, since holding the true band share at zero and drawing the latent effects from a t with five and with three degrees of freedom, and spreading the recipe variances sixteen-fold, leaves the rejection rate between 0.034 and 0.058 against a nominal one in twenty. The shape run used seed 20260968 at 4,000 replicates and the permutation run used seed 20260969 at 500 replicates with 999 permutations per cell.

We then asked the data how much of that effect it actually holds. Lining each configuration's runs up by their batch slot and averaging the run deviations over the 25 recipes of a band leaves a band-shared component untouched while shrinking anything specific to a recipe. The weighted cross-half covariance of those band means measures the component directly. Permuting each configuration's runs among its own batch slots gives the exact null, because it preserves every configuration's own deviations and destroys only the alignment across recipes. Over 2,000 permutations the observed statistic sits inside the null on all four batteries, with 0.28 and 0.74 of permutations at or above it on the full battery and 0.13 and 0.36 without BoolQ. The permutation mean matches the independence prediction to within 1.3 percent everywhere, which lets us read the statistic as a share of the seed covariance. That share comes to 0.008 on full-battery margins with a 95 percent upper bound of 0.044, and to 0.020 without BoolQ with an upper bound of 0.053.

A bound is only worth reading if the test behind it can see the thing it bounds. We measured that on simulated populations whose band share we set, at 500 replicates and 999 permutations each. At a true share of zero the test rejects 0.044 of the time at one in twenty and 0.014 at one in a hundred, so it carries the size it advertises. The plug-in share estimate misses the truth by 0.0009 or less up to the bound itself and by 0.0074 or less at every share through a quarter. Its power climbs from 0.218 at a share of 0.02 to 0.490 at 0.044, then 0.838 at 0.10 and 0.992 at a quarter. The test would almost always catch the share that costs twelve points of coverage, while a share sitting at the bound itself would escape it about half the time.

That measurement carried the margin battery's item noise. The paper reports the same bound on the two accuracy batteries, where the noise doubles. Repeating the sweep there keeps the test honest about its size, since it rejects 0.048 of the time at one in twenty and 0.006 at one in a hundred when the true share is zero. Its plug-in estimate misses the truth by 0.0034 or less at every share up to a quarter. The accuracy scale costs power instead, because the test rejects 0.110 of the time at a share of 0.02, then 0.308 at the 0.044 bound against 0.490 on margins, 0.602 at a tenth against 0.838, and 0.950 at a quarter against 0.992. The reported standard error understates the spread of the estimate there in the same way, at 0.0391 against an actual 0.0571 at the bound and 0.0400 against 0.1199 at a quarter, while the two sit at 0.0395 and 0.0415 when no sharing is present. A reader should therefore treat the accuracy bound as the weaker of the two, because a band share sitting at 0.044 escapes this test in about seven replicates out of ten.

That run used seed 20260990 at 500 replicates with 999 permutations per cell. The standard error we quote is the standard deviation of the permutation null, and that understates the spread of the estimate once the true share is large, at 0.0216 against an actual 0.0386 at a share of 0.044 and 0.0219 against 0.0942 at a quarter. The bound is therefore trustworthy near zero, where the observed estimate of 0.008 sits and where the interval around it covers 0.962. It runs anti-conservative out at the larger shares it is meant to exclude. We read it as a limit on what these data can resolve rather than as a confidence statement about the truth.

Because the smallest share in that first sweep was 0.25, we repeated the sweep at 0.02, 0.05 and 0.10 with 4,000 replicates each, which brackets the estimate and its bound and reaches twice the bound. The recipe-clustered wild interval covers 0.946, 0.941 and 0.929 across those three, with a Monte Carlo standard error near 0.004, so at the bound itself the interval loses about half a point of coverage rather than the twelve points it loses at a quarter. The mirror test, which averages over the five sizes of a recipe instead, also puts the observed statistic inside its null on all four batteries, with tail shares from 0.086 to 0.556. That version is far weaker, because five configurations per recipe leave a standard error of 0.067 on the share against 0.018 for the band version. It bounds recipe-level sharing only below 0.114 on full-battery margins. Weaker doesn't mean blind, and we measured its power the same way we measured the band version's, sweeping the recipe share over 0, 0.05, 0.114, 0.25 and 0.50 at 500 replicates and 999 permutations each. The mirror test rejects 0.054 of the time when the true share is zero, its plug-in estimate misses the truth by 0.0046 or less at every share. Its power runs 0.248 at 0.05, 0.628 at its own 0.114 bound and 0.982 at a quarter. The bound the

corner simulation leans on is therefore a measurement rather than an artefact, since a recipe share of a quarter would be caught in 98 replicates out of 100.

Neither the band test nor the mirror test speaks to the corner where both channels sit at their bounds together. We simulated that corner as well, at 4,000 replicates per cell. Recipe sharing alone at 0.114 leaves the recipe-clustered interval at 0.949, and twice both bounds takes it to 0.927. The two cells that carry the limitation are band sharing alone at its 0.044 bound and the joint corner where both channels sit at their bounds together. The first pass separated those by 0.012, at 0.937 against 0.949, which is a gap 4,000 replicates can't resolve. We repeated both at 12,000 replicates with a fresh seed, and the separation doesn't survive. The larger run puts band sharing alone at 0.9444 and the joint corner at 0.9428, half a Monte Carlo standard error apart, with median widths of 0.242 and 0.246. Pooling the two passes over 16,000 replicates gives 0.9425 and 0.9444. Both corners therefore cost the interval between half a point and a point of coverage. The earlier reading that the two sharing channels partly cancel rested on a difference the first pass was too small to measure. A remedy recovers even that small loss, and it costs almost nothing. Scaling each recipe's residual by one over one minus its leverage, which is the cr3 correction, lifts the recipe-clustered interval from 0.942 to 0.950 at the band corner and from 0.946 to 0.954 at the joint corner over 12,000 replicates, and it widens the median interval only from 0.241 to 0.252. We report the uncorrected interval in the main text because that is what the shipped estimator computes. A reader who wants the corner protected can buy it for four percent of width.

That corner argument was measured on margins, and the paper reports an accuracy conclusion as well. We repeated the same five cells at the accuracy battery's correlation and item noise. The recipe-clustered interval covers 0.950 with no sharing, 0.942 with band sharing alone at its bound, 0.951 with recipe sharing alone at its bound, 0.946 at the joint corner and 0.940 at twice both bounds, with no undefined replicate anywhere until the doubled cell. The corner therefore costs the accuracy interval the same half point to a point it costs the margin one. The limitation transfers between the two scales rather than belonging to one of them. The size-clustered fallback behaves here exactly as the sharing sweeps found, covering between 0.822 and 0.844 with a ninth to an eighth of replicates undefined, while its cluster-robust counterpart holds between 0.930 and 0.943 at every cell including the corner (seed 20260988, 4,000 replicates). The remedy transfers to that scale as well, and it overshoots a little. At 12,000 replicates the leverage correction lifts the recipe-clustered interval from 0.944 to 0.954 at the joint corner and from 0.950 to 0.959 at the band bound, while with no sharing present it moves 0.949 to 0.956. It buys back a point where a point is missing and adds one where none is. Its cost is five percent of width, from 0.395 to 0.414, which sits close to the four percent it costs on margins. A reader shouldn't carry the same correction to the size-clustered fallback, because there it takes coverage from 0.841 down to 0.751 and raises the undefined share from 0.119 to 0.229, since five clusters leave the leverage term almost nothing to work with (seed 20260992, 12,000 replicates). The sharing our clustering assumes is therefore a precaution that these data can neither confirm nor rule out, while the sharing that would break that clustering is bounded where it costs under a point of coverage.

Registered held-out test The protocol fixed the task list, the replacement rule, the statistic, the interval, the pass rule, and a scope rule before any production score existed, and it records every later change as a dated amendment. The OLMES harness at a pinned commit builds every prompt and continuation, MedMCQA contributes 1,000 items drawn under a fixed seed, and DROP and CoQA are each capped at 1,000 items. Halves come from the clustered split of the passage-aware check with a seed fixed in the protocol, so items that share a passage or a story never straddle halves. The CoQA story grouping gives 479 groups for its 1,000 items. Before production, the six pilot checkpoints of one recipe at 150M and 1B each matched their own release seed on 300 ARC-Easy items. The backup rule then read the pilot's three 1B runs, where LogiQA-en gave mean accuracy 0.269 against a bar of 0.270 and LSAT-LR gave 0.219 against 0.220, with chance at 0.25 and 0.20. Both fell short, so the rule replaced them in its listed order with DROP and CoQA as multiple choice, and it reads accuracy only. Each miss is smaller than the pilot's binomial standard error of about 0.010 over three runs and 651 or 510 items. The rule rather than the evidence therefore decided the swap, and we applied it as written. Before each run was scored, a spread of its items was checked against a one-sequence reference path to within 0.001 nats, with a slower agreeing path as the fallback. Pilot timing projected 66.0 session hours for all 375 runs and 54.9 for the 225 runs in scope, which triggered the scope rule. That budget also led us to withdraw a six-task secondary analysis that we had registered earlier.

The request file changed twice after the freeze. Neither change touched a prompt, continuation, label, or item order. The first replaced a salted hash in the passage-group labels with a deterministic one. The second regrouped CoQA by story once it became a primary task. That second change rewrote the group field on 932 rows. Two production shards were scored on the file before the CoQA regrouping. The analysis reads group labels only from the final file, since scores are keyed by task, document, and choice. All 225 runs in scope were scored, 205 of them on Kaggle T4 cards and the last 20 at 1B on three RTX 3090 cards with torch 2.14.0 in place of the T4 image's 2.10.0 after the weekly quota ran out, with the same scorer, float32 weights, 6,000-token cap, and pinned transformers 4.57.1, and every one of those 20 runs passed the 0.001-nat batch check below 0.0001. The pilot's three c4 runs at 1B were scored before either change to the file. The test in Section 4.2 reads all three sizes, and three earlier results preceded it. An earlier interim on the 530M runs alone gave held-out margin inflation of 1.198 with interval [0.864, 1.454] and accuracy inflation of 1.373 with interval [1.138, 1.567], which failed the rule as well. The two finished sizes then gave held-out margin inflation of 1.201 with interval [0.546, 1.616] and accuracy inflation of 1.327 with interval [1.153, 1.484], which failed the rule at a scope where the original ten benchmarks also failed it, at 1.316 with interval [0.880, 1.637]. A second interim, logged in the protocol before either build ran, adds the 18 recipes that the finished shards had scored at 1B to the two finished sizes. On those 68 configurations, held-out margin inflation is 1.181 with interval [0.685, 1.552] and accuracy inflation is 1.262 with interval [1.084, 1.422], which fails the rule there too, while the original ten benchmarks on the same configurations give 1.174 with interval [0.954, 1.360]. At the 18 scored 1B configurations alone, the held-out estimate is 1.152 with interval [0.599, 1.516], and the original ten give 0.947 with interval [0.295, 1.306]. The seven recipes missing at 1B sit in the last shard, which means shard order rather than any 1B number fixed the interim's recipe set. The analysis kernel first recomputes the shipped margin estimate from the 375 released runs and stops unless it returns 1.24395 to within 0.0005, which it did. The 60 four-benchmark subsets at 530M and 750M come from the battery-size kernel run at that scope before any 750M held-out score existed, and 30 of them fell below 1.201. The all-size subsets in Table 2 come from the curve the protocol committed to, and 32 of their 60 fall below the full-scope 1.217. A three-size curve run after that result, and so post hoc, gives a four-benchmark median of 1.172 with 5th and 95th percentiles of 0.979 and 1.485, with 32 of its 60 subsets below 1.217, the same count as at all sizes, and 0.55 of them excluding one against 0.57 there.

Table 20 removes one held-out task at a time. Pooling the four held-out tasks with the original ten gives margin inflation of 1.419 with interval [1.051, 1.714] and accuracy inflation of 1.240 with interval [0.986, 1.458]. Table 21 gives the battery-size curve at 530M, 750M, and 1B, at 530M and 750M, and at 530M alone alongside the all-size curve that the protocol committed to reporting. Every entry reads subsets of the original ten benchmarks on the released runs, so none of them uses a held-out score.

The held-out tasks carry 1,000 items each, while the original benchmarks range from 500 to 14,042. A separate kernel asked what item count costs the interval. For each cap it drew that many items per benchmark under a fixed seed, split them into halves by passage group, and recomputed every run's trait scores on the two halves. The uncapped entry therefore uses the kernel's own split rather than the primary one. Capping at 1,000 items per benchmark leaves 9,500 of the 37,682 items and gives full-battery margin inflation of 1.218 with interval [1.121, 1.309], against 1.247 with interval [1.146, 1.342] uncapped. A cap of 200 gives 1.179 with interval [1.002, 1.337]. Over 20 sampled four-benchmark subsets, the margin share whose interval excludes one stays between 0.45 and 0.55 at every cap from 200 items to the full counts. Item count therefore doesn't explain the four-task failure rate. At the registered scope of 530M, 750M, and 1B, the original ten benchmarks give margin inflation of 1.240 with interval [1.054, 1.397] and accuracy inflation of 1.097 with interval [0.922, 1.246]. We computed both from the released original-benchmark scores inside the kernel that reported the test.

Table 19 gives the rule's simulated pass rate at each scope it could take. The item noise is 0.73 at the margin-like level, 1.45 at the accuracy-like level, and 1.00 between them. We fixed that design before the two-size analysis ran. At a true ratio of one the pass rate stays between 0.011 and 0.022 at the three fixed noise levels, so the rule holds its nominal size. At two sizes and a true ratio of 1.244, the median simulated interval width is 0.318 at margin-like noise and 0.530 at accuracy-like noise, against the observed 1.070. After seeing that width we added noise levels of 2.00, 2.50, and 3.00, whose two-size median widths at a true 1.244 are 0.765, 0.972, and 1.146. At 2.50 and 3.00 the pass

rate at a true 1.244 is 0.115 and 0.080 against a size of 0.024 and 0.025. The wild interval's coverage there falls to 0.699 and 0.562, and 0.280 and 0.425 of the intervals have an undefined lower limit, the same failure that the tables above show on the data. At three sizes the median widths at a true 1.244 are 0.258, 0.306, and 0.431 at the three fixed noise levels and 0.640, 0.839, and 1.009 at the added ones, against the observed 0.626, and at 2.00 the pass rate is 0.262 with coverage 0.918 and 0.040 of intervals undefined. An exploratory run on the 750M runs alone gives held-out margin inflation of 1.203 with an undefined lower limit and an upper limit of 1.828, and accuracy inflation of 1.299 with interval [0.959, 1.584]. The original ten benchmarks give 1.110 with interval [0.686, 1.407] and 0.987 with interval [0.310, 1.320] at that size.

Table 19. Simulated pass rate of the registered rule, a lower 95% wild bootstrap limit above one, by scope, item-noise level, and true held-out inflation, from a compound-symmetric generator with four traits, 25 recipe clusters, and three runs. Each cell uses 4,000 replicates, so its Monte Carlo error is at most 0.008. We fixed the design at noise 0.73, 1.00, and 1.45 before the 530M and 750M analysis ran, and added the rows at 2.00, 2.50, and 3.00 after seeing the observed interval width.

Scope, configurations	Noise	1.00	1.10	1.20	1.244	1.316	1.40	1.50
One size, 25	0.73	0.011	0.088	0.318	0.454	0.681	0.884	0.974
Two sizes, 50	0.73	0.016	0.180	0.600	0.785	0.945	0.995	1.000
Three sizes, 75	0.73	0.019	0.258	0.792	0.926	0.992	1.000	1.000
One size, 25	1.00	0.015	0.074	0.238	0.329	0.535	0.752	0.905
Two sizes, 50	1.00	0.015	0.138	0.454	0.631	0.847	0.961	0.998
Three sizes, 75	1.00	0.019	0.189	0.640	0.820	0.963	0.997	1.000
One size, 25	1.45	0.020	0.057	0.142	0.192	0.324	0.488	0.683
Two sizes, 50	1.45	0.017	0.083	0.259	0.371	0.581	0.794	0.943
Three sizes, 75	1.45	0.022	0.110	0.368	0.524	0.754	0.923	0.991
One size, 25	2.00	0.028	0.048	0.099	0.118	0.185	0.281	0.411
Two sizes, 50	2.00	0.023	0.060	0.141	0.188	0.322	0.477	0.686
Three sizes, 75	2.00	0.025	0.070	0.195	0.262	0.436	0.634	0.824
One size, 25	2.50	0.027	0.039	0.073	0.083	0.119	0.168	0.242
Two sizes, 50	2.50	0.024	0.050	0.092	0.115	0.189	0.280	0.438
Three sizes, 75	2.50	0.028	0.053	0.123	0.162	0.255	0.381	0.548
One size, 25	3.00	0.028	0.034	0.053	0.065	0.081	0.110	0.147
Two sizes, 50	3.00	0.025	0.041	0.067	0.080	0.119	0.161	0.259
Three sizes, 75	3.00	0.026	0.046	0.085	0.106	0.154	0.223	0.335

Table 20. Held-out estimates at 530M, 750M, and 1B with one task removed, with nominal 95% recipe-cluster intervals over 75 configurations.

Task removed	Margin	Interval	Accuracy	Interval
SciQ	1.120	[0.800, 1.393]	1.055	[0.909, 1.187]
MedMCQA	1.157	[0.836, 1.426]	1.252	[1.107, 1.380]
DROP	1.213	[1.035, 1.377]	1.128	[0.998, 1.247]
CoQA	1.090	[0.990, 1.181]	1.118	[0.848, 1.341]

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Table 21. Median margin inflation over subsets of the original ten benchmarks, with the share of subsets whose nominal 95% interval excludes one, at 530M and 750M, at 530M, 750M, and 1B, at 530M alone, and across all five sizes. Benchmark counts with more than 60 subsets use 60 sampled without replacement, and subsets whose bootstrap degenerates are left out of that share. The two-size rows were fixed before any 750M held-out score existed, the 530M rows came after the 530M interim result, and the three-size rows came after the full-scope result. These intervals use 1,999 wild draws, against 4,999 for the registered test.

Benchmarks	2	3	4	5	6	7	8	9	10
530M and 750M median	1.010	1.143	1.213	1.288	1.142	1.206	1.210	1.259	1.316
530M and 750M share	0.37	0.53	0.55	0.53	0.37	0.33	0.20	0.10	0.00
530M, 750M, and 1B median	1.014	1.109	1.172	1.227	1.097	1.156	1.163	1.196	1.240
530M, 750M, and 1B share	0.36	0.48	0.55	0.50	0.42	0.45	0.58	0.80	1.00
530M median	1.019	1.133	1.244	1.317	1.296	1.368	1.377	1.439	1.514
530M share	0.33	0.58	0.67	0.80	0.78	0.87	0.91	0.90	1.00
All sizes median	1.012	1.116	1.176	1.228	1.100	1.160	1.160	1.200	1.244
All sizes share	0.38	0.53	0.57	0.62	0.73	0.82	0.98	1.00	1.00

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